

The Factors Behind Life

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Introduction

"Life", a four-letter word that carries one of the heaviest meanings for all of us. People aspire to have a good life, a happy life, and often a long life. Since "life" is such an important topic, it is natural to ask whether we can predict a certain aspect of "life" through statistical means or not. Is this process possible? In this project, we explore this aspect of life through the lens of regression analysis.

Our dataset contains socioeconomic, health-related, and demographic variables from countries around the world. Our goal is to examine life expectancy and investigate any interesting relationships between life expectancy and the variables from our dataset. Additionally, we hope to answer these questions for our project: Do the relationship between life expectancy and the selected variables make sense? From our regression model, what is the biggest factor influencing "life expectancy"? How accurate are our predictions from this regression model? By exploring these questions, we hope to gain insight into what shapes human longevity and better understand our "life".

Multicollinearity

We want to narrow down the list of variables to those that are most relevant for our regression analysis. We want 'life_expectancy' to be our response variable, so we keep that column and select predictor variables that are likely to have an impact on life expectancy. First, we will remove all non-numeric variables and only consider those that are numeric. Then we want to assess the correlation amongst all remaining regressors and keep those that are loosely correlated with 'life_expectancy' to narrow down our list of predictors. We want to use a less

stringent correlation threshold of $|r| > 0.10$ to keep more potential predictors for our regression model. After looking at the correlation matrix for `life\_expectancy` we keep the following variables as predictors: density_pop, agri_land_pct, birth_rate, co2, cpi, cpi_pct_change, fertility_rate, gas_price, gdp, primary_enroll_pct, tertiary_enroll_pct, infant_mortality, mmr, min_wage, oop_health_exp, phys_per_1k, labor_force_pct, tax_rev_pct, and total_tax_rate. From the pairwise plot, we can see that some of the predictors have non-linear relationships with life_expectancy. We may need to consider transforming some of these variables or using polynomial terms in our regression model to better capture these relationships. In particular, density_pop, co2, cpi, cpi_pct_change, gdp, min_wage, and phys_per_1k show strong non-linear patterns with life_expectancy. Looking at the correlation matrix again, we can see that some of the transformations have improved the linearity of the relationships between predictors and life_expectancy. We also observe multiple pairwise collinearity amongst some predictors. Notably, birth_rate, infant_mortality, and fertility_rate are all highly positively correlated with each other. Also gdp and co2. We will need to consider removing one or more of these variables from the model to reduce multicollinearity before variable selection. We will be using variance inflation factor (VIF), examining condition indices, and looking at variance decomposition proportions to assess multicollinearity in our regression model. From the VIF, we can see that birth_rate, co2, fertility_rate, gdp, and infant_mortality have VIF values > 10 , indicating multicollinearity. These are the exact variables identified earlier as being highly correlated with each other via the correlation matrix and pairwise plots. We also look at condition indices and variance decomposition proportions to further assess multicollinearity. We see that both `co2` and `gdp` have the highest variance decomposition proportions for the same high condition index of 5395.4706, indicating this pair is contributing to multicollinearity. We also saw `gdp` had a

VIF > 10 so we will remove `gdp` from the model first and continue reassessing multicollinearity. Assessing VIF values after removal of `gdp`, we can see that birth_rate, fertility_rate, and infant_mortality still have VIF values > 10 indicating multicollinearity. Looking at the variance decomposition proportions one more time, we see that birth_rate and fertility_rate have high variance decomposition proportion for the high condition index of row 15 ($85.2164 > 30$), indicating it is contributing to multicollinearity. We remove `birth\_rate` from the model next and continue reassessing multicollinearity. CI and VP no longer suggest multicollinearity, however we still see that `infant\_mortality` with VIF > 10 even after removal of `gdp` and `birth\_rate`. This suggests that we have not successfully reduced multicollinearity in our model. We'll note that infant_mortality, fertility_rate, phys_per_1k, and mmr, and still showing correlation ($|r| > 0.8$) among pairwise combinations of each. This is also supported if we choose a more restrictive threshold for VIF like 5.

Looking into the issue more in depth, the majority of this collinearity is caused by `infant\_mortality`. Life expectancy in this dataset is defined as "the average number of years a newborn is expected to live". As the following article explains (<https://ourworldindata.org/life-expectancy-how-is-it-calculated-and-how-should-it-be-interpreted>), the life table—which generates life expectancy—integrates mortality across the entire lifespan, starting from birth. Since infant mortality is a direct component of life expectancy at birth. High infant mortality lowers life expectancy at birth; low infant mortality raises it. This relationship is deterministic and is built into the definition and calculation of `life\_expectancy`. Including infant mortality as a predictor treats it as if it were an independent regressor, when in fact it is a component of the response variable itself. It is possible for any given population to have a low life expectancy at birth and yet have a large proportion of old people. Including infant mortality as a predictor

could falsely inflate our predictive power and fail to capture real relationships with predictors associated with lifestyle choices, because infant mortality is a component of the response variable itself. This led us to remove `life\_expectancy` from our model. Similarly `mmr` - maternal mortality rate and `fertility\_rate` introduces the same issues. Therefore, we will also remove infant_mortality, mmr, and fertility_rate from the model to avoid this issue. We look at VIF once last time and see all remaining variables have $VIF < 5$. Now that we have reduced multicollinearity in our model, we can proceed with variable selection. We will use stepwise selection based on AIC and regsubsets() to identify the best subset of predictors for our regression model.

Variable Selection and Final Model Specification

With the dataset fully cleaned, missing values appropriately handled, and severe multicollinearity eliminated through the removal of redundant proxies (e.g., GDP, infant mortality, fertility rate, and maternal mortality ratio), the analysis could proceed to the critical stage of variable selection. The objective at this point was clear: from the remaining theoretically plausible and empirically distinct predictors, identify the subset that best explains variation in life expectancy while maintaining parsimony, interpretability, and genuine predictive performance.

Stepwise Regression Using AIC

Variable selection began with bidirectional stepwise regression based on Akaike's Information Criterion (AIC). This procedure iteratively adds and removes predictors from a saturated model, at each step retaining the change that produces the largest improvement in AIC until no further

improvement is possible. The algorithm converged on a model containing nine predictors with a final AIC of 313.80—a substantial reduction from the full model.

Validation via Exhaustive Best-Subsets Regression

To guard against the possibility that stepwise selection had settled on a locally rather than globally optimal solution, exhaustive best-subsets regression was performed using the `regsubsets()` function from the `leaps` package. This computationally intensive approach evaluates every possible subset of the candidate predictors and identifies the single best model of each size. Results confirmed the stepwise findings: the globally optimal model under AIC also contained exactly nine predictors—the identical set selected earlier—and achieved a marginally lower AIC of 311.80.

Additional Selection Criteria

Further examination of adjusted R^2 and Mallows' C_p reinforced the preference for a nine-predictor specification. Adjusted R^2 reached its maximum value of 0.777 at nine predictors before declining, while Mallows' C_p was closest to its ideal value ($C_p \approx p$) at the same model size. In contrast, the Bayesian Information Criterion (BIC), which imposes a heavier penalty for additional parameters favored a more parsimonious five-predictor model.

Predictive Performance via Cross-Validation

To resolve the tension between criteria favoring complexity (AIC, adjusted R^2 , C_p) and the stricter BIC, 10-fold cross-validation was conducted using the `caret` package. The nine-predictor model yielded a cross-validated RMSE of 3.571 years, compared to 3.602 years for the five-predictor

specification a difference of only 0.031 years, which is of negligible practical importance in the context of country-level life expectancy.

Final Model Selection

Given the trivial predictive advantage of the larger model, the principle of parsimony was applied. The five-predictor specification strongly supported by BIC and offering virtually identical out-of-sample performance was therefore adopted as the final model:

```
best_model <- lm(life_expectancy ~ density_pop + agri_land_pct +  
  tertiary_enroll_pct + min_wage + phys_per_1k,  
  data = world_model_data)
```

Results of the Final Model

The selected model is highly significant overall ($F(5, 116) = 76.19, p < 2.2 \times 10^{-16}$) and explains 76.7 % of the global variation in life expectancy (adjusted $R^2 = 0.756$). All five predictors are statistically significant at conventional levels, and their estimated effects are substantively meaningful:

Diagnostic Checks

Residual diagnostics revealed no serious violations of regression assumptions. The residuals versus fitted plot shows random scatter around zero with no pronounced patterns or heteroscedasticity. A slight downward tilt in the loess smoother is noted but remains minor and does not compromise the validity of inference.

Influential Observation

After selecting our final model through variable selections, we assessed whether any observations would be influential for our regression results because if they are, they could make our model biased towards certain observations. To assess the influential observations, we sent a half-normal plot, studentized residuals, hat values, and Cook's distance. The half-normal plot helped me visualize the hat values and the Cook's distance that shows if the outliers are influential or not. While the half normal plot that visualizes studentized residuals shows the potential outliers and doesn't tell us about its influence. These three values will be used hand-to-hand to pinpoint potential outliers and their influence on the plot. Therefore, we decided to use `influencePlot()` to create a table and a plot that shows student residuals, hat values, and cook's distance all in one plot to determine influential observations.

To interpret these diagnostics, we used common thresholds for interpreting student residuals, hat values, and Cook's distance. Observations with unusually high studentized residuals when compared to other observations would mean that the observation is a potential outlier, but these alone don't indicate influence. High leverage points are identified if the value exceeds 0.147 ($3 * \text{predictors/number of observations}$) which corresponds to the observation having an unusually strong pull on the model. However, the leverage value by itself doesn't determine influence, so we need to incorporate Cook's distance into our evaluation. For interpreting the Cook's distance, the observation must exceed 0.5 value for the observation to be considered influential. With these in mind, we will write about the potential influential observations in the next few paragraphs.

The function `influencePlot()` flagged 4 observations that are potential influential observations located in rows 24, 40, 66, and 84. For the observation on row 24, it had a high studentized residual, indicating it was an outlier, but it didn't have high hat value and Cook's distance; therefore, it wasn't considered influential. For the observation on row 40, it didn't have high student residual nor Cook's distance, but it had high hat value (exceeding 0.147). However, due to the low studentized residual, low Cook's distance, and how the hat value was not too much higher than the threshold, this observation was also deemed non-influential.

For observation on row 66, it had high studentized residuals, and a slightly elevated hat value with low Cook's distance and was likewise not considered to be an influential observation.

Lastly, for the observation on row 84, it had high studentized residuals, but both the hat value and Cook's distance were low. Consequently, none of these potentially influential observations were removed from the dataset.

Checking Error Assumptions

With the influential observations addressed, we proceeded to check the error assumptions before starting our interpretation. Specifically, we are checking if our model satisfies the mean zero assumption, the constant variance assumption, the normality assumption, and the uncorrelated errors assumption. These checks were performed using residuals vs. fitted plots, scale-location plot, non-constant variance test, Q-Q plot, histogram plot, lag plot of residuals, and the Durbin-Watson test.

To check the mean-zero assumption, we examined a residuals vs fitted plot. The red line had a weird pattern and deviated from zero, suggesting that the residuals didn't have a mean of zero.

To investigate further, we examined the residual plots of the predictors vs. the fitted values and

found patterns in the “tertiary enrollment percentage” and “physicians per 1000 people” variables. This indicated that these variables needed a transformation. Therefore, we used a natural log to transform both of these variables which improved the residuals vs. fitted plot: the red line was now closer to the mean-zero line with no patterns. After this transformation, the mean-zero assumption was satisfied.

To assess the constant variance assumption, we created a scale-location plot and conducted a non-constant variance score test. From the scale-location plot, we don't see a pattern in the red line, and we also noticed that the residuals seem somewhat uneven. To verify, we ran the non-constant variance score test, which evaluates whether the variance of residuals depends on the fitted values or not. The test evaluates the dependency of residuals by performing null & alternative hypotheses tests where the null hypothesis states that the model has constant variance and the alternative hypothesis states otherwise. From the test, we got high chi-square values and an extremely low p-value which states strong evidence to reject the null hypothesis. Therefore, this test concludes that our model does not satisfy the constant variance error assumption.

To check the normality error assumption, we examined a Q-Q plot and a histogram of the residuals. The Q-Q plot showed slight steepness at the left tail, but the majority of the points followed closely with the reference line which suggests that the error has an approximately normal distribution. To verify this claim, we also made a histogram to see if the distribution is approximately normal. As a result, the histogram distribution's overall shape resembles a bell curve even though it's not perfect. Therefore, I believe the normality assumption is satisfied.

To assess the uncorrelated error assumption, we used a lag plot of residuals and a Durbin-Watson test to check for correlation. The lag plot of residuals displayed a circular cloud of points with no

direction, trend, or clustering which suggests little to no correlation. To verify this claim, we conducted a Durbin-Watson test. The Durbin-Watson test provides a more formal evaluation: a statistic approximately to 2 indicates no correlation. In addition, the function for performing the Durbin-Watson test also provides us with a hypothesis test where the null hypothesis corresponds to having no correlation and the alternative hypothesis states otherwise. Our model produced a Durbin-Watson score of approximately 2 and a p-value of 0.5386 which tells us that our model probably has uncorrelated errors and that we don't have statistically significant evidence to reject the null hypothesis. Together, these results indicate that the uncorrelated error assumption is satisfied.

Interpretation and Prediction

Now that we have our model, we interpret the predictors and how they relate to our response variable.

$$\widehat{\text{Life_Exp}} = 60.5 + 0.885(\text{density_pop}) - 0.908(\text{agri_land_pct}) + 3.31(\text{min_wage}) + 1.62(\text{phys_per_1k}) + 2.58(\text{tertiary_enroll_pct})$$

- Intercept: When all other regressors are equal to zero, the average predicted life expectancy is 60.5 years.
- density_pop: When the population density in persons per square kilometer increases by a factor of 10 while other regressors remain constant, the average predicted life expectancy will go up by 0.885 years.
- agri_land_pct: When the percentage of land area used for agricultural purposes increases by a factor of 10 while other regressors remain constant, the average predicted life expectancy will go down by 0.908 years.

- `min_wage`: When the minimum wage level in local currency increases by a factor of 10 while other regressors remain constant, the average predicted life expectancy will go up by 3.31 years.
- `phys_per_1k`: When the number of physicians per thousand people increases by a factor of 10 while other regressors remain constant, the average predicted life expectancy will go up by 1.62 years.
- `tertiary_enroll_pct`: When the gross enrollment ratio for tertiary education increases by a factor of 10 while other regressors remain constant, the average predicted life expectancy will go up by 2.58 years.

Overall, we can see that higher population density, minimum wage, number of physicians per thousand people, and tertiary education enrollment are associated with higher life expectancy.

On the other hand, a higher percentage of agricultural land is associated with lower life expectancy. These relationships make sense as they reflect factors that can influence access to healthcare, economic stability, and educational opportunities, all of which can impact overall health and longevity.

The predictions in our model give us the estimated life expectancy for the new data points along with the 95% prediction intervals. These intervals give us a range within which we can expect the true-life expectancy to fall for each new observation, accounting for both the uncertainty in the model estimates and the inherent variability in life expectancy.

From the predictions, we can see that as population density, minimum wage, number of physicians per thousand people, and tertiary education enrollment increase, the predicted life expectancy also increases. Conversely, as the percentage of agricultural land increases, the predicted life expectancy decreases. This aligns with our earlier interpretations of the regression

coefficients. Overall, our regression model provides valuable insights into the factors influencing life expectancy and allows us to make informed predictions for new data points based on these relationships.

Conclusion

Our regression analysis has shed light on independent key factors that influence life expectancy across different countries. We found that higher population density, minimum wage, number of physicians per thousand people, and tertiary education enrollment are positively associated with life expectancy, while a higher percentage of agricultural land is negatively associated.

One of the error assumptions was violated. Specifically, the constant variance assumption. This could be caused by omitted variable bias, where there are important predictors that were not included in the model. Future work could explore keeping relevant variables, such as `fertility\_rate` to help address this issue. Adding interaction terms or polynomial terms for some predictors to better capture non-linear relationships could also help with this issue. Alternatively, the non-constant variance could be explained by the response itself. As you get older, life expectancy is less variable.

Our final regression model demonstrated good predictive capabilities, allowing us to estimate life expectancy for new data points with an adjusted R squared = 0.8029. The 95% prediction intervals provided a useful measure of uncertainty around these estimates, emphasizing the non-constant variance observed in our error assumption analysis.

Overall, this project has provided valuable insights into the complex interplay of factors that shape human longevity. By understanding these relationships, policymakers and public health officials can better target interventions to improve life expectancy and overall well-being in

populations around the world. Further research could explore additional variables and longitudinal data to deepen our understanding of the determinants of life expectancy.

Appendix

Imports

Importing Packages

```
{r}
library(tidyverse)
library(janitor)
library(car)
library(mctest)
library(lmtest)
library(leaps)
library(caret)
library(faraway)
```

Importing Data

```
{r}
# Import the data
world <- read_csv("~/Downloads/world-data-2023.csv")

# Check the first 6 rows of the dataset
head(world)
```

A tibble: 6 × 35

Country <chr>	Density (P/Km2) <dbl>	Abbreviation <chr>
Afghanistan	60	AF
Albania	105	AL
Algeria	18	DZ
Andorra	164	AD
Angola	26	AO
Antigua and Barbuda	223	AG

6 rows | 1-3 of 35 columns

Data Cleaning

Cleaning Column Names

```
{r}
# Use clean_names() from janitor to clean the column names
world_clean <- clean_names(world)

# Check the new column names
head(world_clean)
```

A tibble: 6 × 35

country <chr>	density_p_km2 <dbl>	abbreviation <chr>
Afghanistan	60	AF
Albania	105	AL
Algeria	18	DZ
Andorra	164	AD
Angola	26	AO
Antigua and Barbuda	223	AC

6 rows | 1-3 of 35 columns

Cleaning Column Names (2)

```
{r}
# Rename the columns
world_cleaned <- world_clean |> rename(density_pop = density_p_km2,
                                       abbrev = abbreviation,
                                       agri_land_pct = agricultural_land_percent,
                                       area_km2 = land_area_km2,
                                       forces_size = armed_forces_size,
                                       capital = capital_major_city,
                                       co2 = co2_emissions,
                                       cpi_pct_change = cpi_change_percent,
                                       forest_pct = forested_area_percent,
                                       gas_price = gasoline_price,
                                       primary_enroll_pct =
gross_primary_education_enrollment_percent,
                                       tertiary_enroll_pct =
gross_tertiary_education_enrollment_percent,
                                       mmr = maternal_mortality_ratio,
                                       min_wage = minimum_wage,
                                       language = official_language,
                                       oop_health_exp =
out_of_pocket_health_expenditure,
                                       phys_per_1k = physicians_per_thousand,
                                       labor_force_pct =
population_labor_force_participation_percent,
                                       tax_rev_pct = tax_revenue_percent)

# Converting percentage and rate columns from col_character() to col_double() types
world_cleaned <- world_cleaned |>
  mutate(across(c(agri_land_pct,
                  primary_enroll_pct,
                  tertiary_enroll_pct,
                  birth_rate,
                  cpi_pct_change,
                  fertility_rate,
                  forest_pct,
                  oop_health_exp,
                  labor_force_pct,
                  tax_rev_pct,
                  total_tax_rate,
                  unemployment_rate,
                  urban_population,
                  population),
                ~ as.double(str_replace(., "%", ""))))
```

Cleaning Column Names (3)

```
# Converting percentage and rate columns from col_character() to col_double() types
world_cleaned <- world_cleaned |>
  mutate(across(c(agri_land_pct,
                  primary_enroll_pct,
                  tertiary_enroll_pct,
                  birth_rate,
                  cpi_pct_change,
                  fertility_rate,
                  forest_pct,
                  oop_health_exp,
                  labor_force_pct,
                  tax_rev_pct,
                  total_tax_rate,
                  unemployment_rate,
                  urban_population,
                  population),
    ~ as.double(str_replace(., "%", ""))))

# Converting gas_price and min_wage columns from col_character() to col_double()
types
world_cleaned <- world_cleaned |>
  mutate(across(c(gas_price,
                  min_wage),
    ~ as.double(str_replace(., "$", ""))))

# Converting gdp column from col_character() to numeric type, removing dollar
symbols and commas
world_cleaned <- world_cleaned |>
  mutate(gdp = as.double(str_replace_all(gdp, "$[,]", "")))
```

Data Exploration

Summary

```
{r}
# Check the summary and column types
summary(world_cleaned)
```

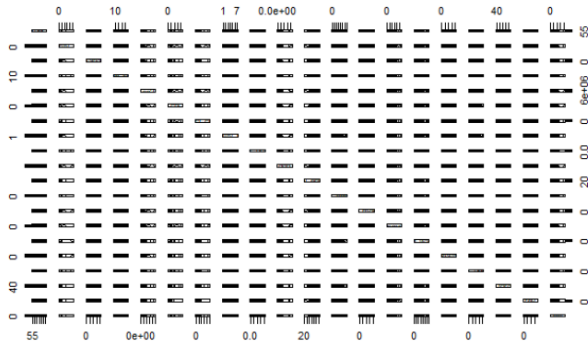
country	density_pop	abbrev	agri_land_pct	area_km2	forces_size	birth_rate	calling_code	capital
Length:195	Min. : 2.0	Length:195	Min. : 0.60	Min. : 0	Min. : 0	Min. : 5.90	Min. : 1.0	Length:195
Class :character	1st Qu.: 35.5	Class :character	1st Qu.:21.70	1st Qu.: 23828	1st Qu.: 11000	1st Qu.:11.30	1st Qu.: 82.5	Class :character
Mode :character	Median : 89.0	Mode :character	Median :39.60	Median : 119511	Median : 31000	Median :17.95	Median : 255.5	Mode :character
	Mean : 356.8		Mean :39.12	Mean : 689624	Mean : 159275	Mean :20.21	Mean : 360.5	
	3rd Qu.: 216.5		3rd Qu.:55.38	3rd Qu.: 524256	3rd Qu.: 142000	3rd Qu.:28.75	3rd Qu.: 506.8	
	Max. :26337.0		Max. :82.60	Max. :17098240	Max. :3031000	Max. :46.08	Max. :1876.0	
			NA's :7	NA's :1	NA's :24	NA's :6	NA's :1	
co2	cpi	cpi_pct_change	currency_code	fertility_rate	forest_pct	gas_price	gdp	primary_enroll_pct
Min. : 11	Min. : 99.03	Min. : -4.300	Length:195	Min. :0.980	Min. : 0.00	Min. :0.000	Min. :4.727e+07	Min. : 23.40
1st Qu.: 2304	1st Qu.:113.89	1st Qu.: 1.000	Class :character	1st Qu.:1.705	1st Qu.:11.00	1st Qu.:0.755	1st Qu.:8.455e+09	1st Qu.: 98.95
Median : 12303	Median :125.34	Median : 2.300	Mode :character	Median :2.245	Median :32.00	Median :0.980	Median :3.439e+10	Median :102.55
Mean : 177799	Mean : 190.46	Mean : 6.722		Mean :2.698	Mean :32.02	Mean :1.002	Mean :4.773e+11	Mean :102.47
3rd Qu.: 63884	3rd Qu.:157.26	3rd Qu.: 4.250		3rd Qu.:3.598	3rd Qu.:48.17	3rd Qu.:1.240	3rd Qu.:2.341e+11	3rd Qu.:108.03
Max. :9893038	Max. :4583.71	Max. :254.900		Max. :6.910	Max. :98.30	Max. :2.000	Max. :2.143e+13	Max. :142.50
NA's :7	NA's :17	NA's :16		NA's :7	NA's :7	NA's :20	NA's :2	NA's :7
tertiary_enroll_pct	infant_mortality	largest_city	life_expectancy	mmr	min_wage	language	oop_health_exp	phys_per_1k
Min. : 0.80	Min. : 1.40	Length:195	Min. :52.80	Min. : 2.0	Min. : 0.0100	Length:195	Min. : 0.20	Min. :0.0100
1st Qu.: 12.15	1st Qu.: 6.00	Class :character	1st Qu.:67.00	1st Qu.: 13.0	1st Qu.: 0.4025	Class :character	1st Qu.:17.45	1st Qu.:0.3325
Median : 31.20	Median :14.00	Mode :character	Median :73.20	Median : 53.0	Median : 1.0450	Mode :character	Median :30.70	Median :1.4600
Mean : 37.96	Mean :21.33		Mean :72.28	Mean :160.4	Mean : 2.1941		Mean :32.64	Mean :1.8398
3rd Qu.: 63.15	3rd Qu.:32.70		3rd Qu.:77.50	3rd Qu.:186.0	3rd Qu.: 2.4475		3rd Qu.:43.98	3rd Qu.:2.9350
Max. :136.60	Max. :84.50		Max. :85.40	Max. :1150.0	Max. :13.5900		Max. :81.60	Max. :8.4200
NA's :12	NA's :6		NA's :8	NA's :14	NA's :45		NA's :7	NA's :7
population	labor_force_pct	tax_rev_pct	total_tax_rate	unemployment_rate	urban_population	latitude	longitude	
Min. :8.360e+02	Min. :38.00	Min. : 0.00	Min. : 8.00	Min. : 0.090	Min. : 5464	Min. : -40.901	Min. : -175.198	
1st Qu.:1.963e+06	1st Qu.:56.15	1st Qu.:11.70	1st Qu.: 30.60	1st Qu.: 3.395	1st Qu.: 1152961	1st Qu.: 4.544	1st Qu.: -7.941	
Median :8.827e+06	Median :62.45	Median :16.30	Median : 37.20	Median : 5.360	Median : 4678104	Median :17.274	Median : 20.973	
Mean :3.938e+07	Mean :62.74	Mean :16.57	Mean : 40.82	Mean : 6.886	Mean : 22304543	Mean :19.092	Mean : 20.232	
3rd Qu.:2.859e+07	3rd Qu.:69.50	3rd Qu.:21.10	3rd Qu.: 48.00	3rd Qu.: 9.490	3rd Qu.: 14903239	3rd Qu.: 40.125	3rd Qu.: 48.282	
Max. :1.398e+09	Max. :86.80	Max. :37.20	Max. :219.60	Max. :28.180	Max. :842933962	Max. : 64.963	Max. :178.065	
NA's :1	NA's :19	NA's :26	NA's :12	NA's :19	NA's :5	NA's :1	NA's :1	

Multicollinearity Checks

```
{r}
world_numeric <- world_cleaned |> dplyr::select(where(is.numeric))
# we also want to remove any columns with location information or identifiers
world_numeric <- world_numeric |> dplyr::select(-c(latitude, longitude, calling_code))
cor_matrix <- cor(world_numeric, use = "pairwise.complete.obs")
cor_matrix[, "life_expectancy"]
```

density_pop	agri_land_pct	area_km2	forces_size	birth_rate	co2	cpi	cpi_pct_change
0.169851798	-0.243771006	0.054921375	0.075943268	-0.874747210	0.121891126	-0.181656290	-0.150087170
fertility_rate	forest_pct	gas_price	gdp	primary_enroll_pct	tertiary_enroll_pct	infant_mortality	life_expectancy
-0.849152811	0.005900126	0.247971117	0.175755336	0.094906544	0.722534733	-0.924675323	1.000000000
mmr	min_wage	oop_health_exp	phys_per_1k	population	labor_force_pct	tax_rev_pct	total_tax_rate
-0.831796516	0.599968610	-0.326161531	0.703740229	0.008815821	-0.154818416	0.363980147	-0.188535432
unemployment_rate	urban_population						
-0.040152334	0.070463068						

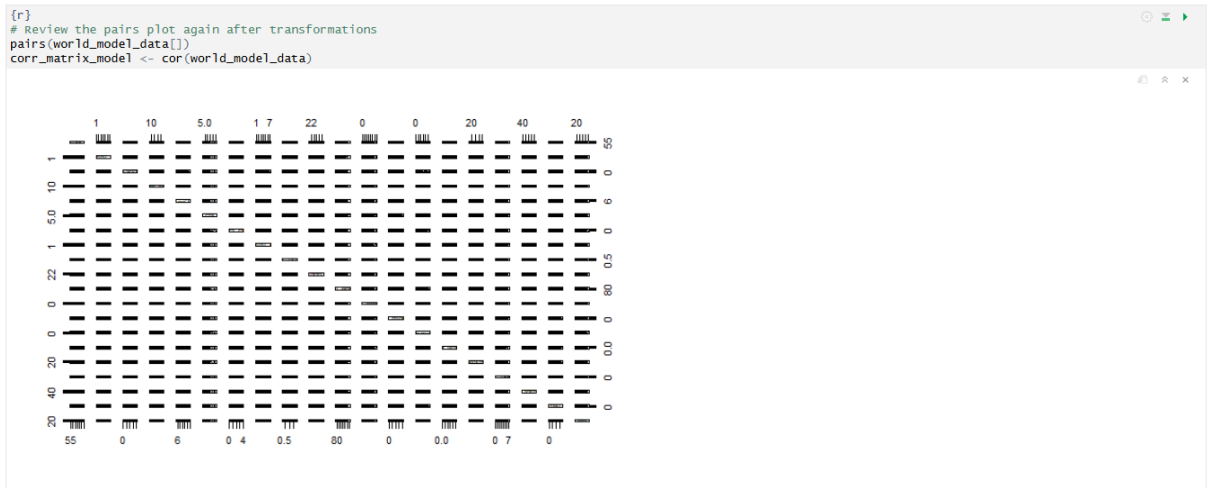
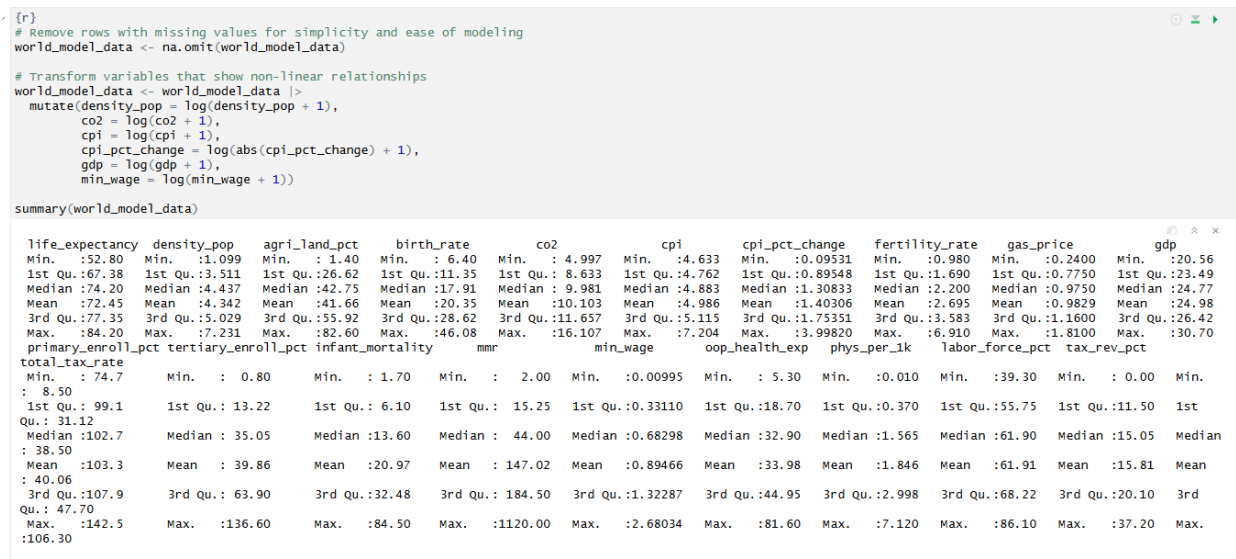
```
{r}
world_model_data <- world_numeric |> dplyr::select(life_expectancy, density_pop, agri_land_pct, birth_rate, co2, cpi, cpi_pct_change, fertility_rate, gas_price, gdp, primary_enroll_pct, tertiary_enroll_pct, infant_mortality, mmr, min_wage, oop_health_exp, phys_per_1k, labor_force_pct, tax_rev_pct, total_tax_rate)
pairs(world_model_data[])
```



```
{r}
# Check for missing values in the selected model data
colSums(is.na(world_model_data))
```

life_expectancy	density_pop	agri_land_pct	birth_rate	co2	cpi	cpi_pct_change	fertility_rate
8	0	7	6	7	17	16	7
gas_price	gdp	primary_enroll_pct	tertiary_enroll_pct	infant_mortality	mmr	min_wage	oop_health_exp
20	2	7	12	6	14	45	7
phys_per_1k	labor_force_pct	tax_rev_pct	total_tax_rate				
7	19	26	12				

Multicollinearity Checks (2)



Multicollinearity Checks (3)

{r}

Fit an initial full model to assess multicollinearity
full_model <- lm(life_expectancy ~ ., data = world_model_data)
summary(full_model)

Call:
lm(formula = life_expectancy ~ ., data = world_model_data)

Residuals:
Min 1Q Median 3Q Max
-5.0271 -1.1404 0.0256 1.1258 4.2871

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	56.588392	7.996493	7.077	1.92e-10 ***
density_pop	0.491814	0.194171	2.533	0.01284 *
agri_land_pct	-0.023179	0.010020	-2.313	0.02271 *
birth_rate	0.118486	0.139890	0.847	0.39898
co2	-0.742590	0.415200	-1.789	0.07666 .
cpi	0.788725	0.866481	0.910	0.36483
cpi_pct_change	-0.439068	0.395201	-1.111	0.26918
fertility_rate	-1.220851	0.957025	-1.276	0.20497
gas_price	-0.394733	0.929456	-0.425	0.67196
gdp	0.867781	0.426357	2.035	0.04441 *
primary_enroll_pct	-0.002110	0.018155	-0.116	0.90771
tertiary_enroll_pct	0.026180	0.013303	1.968	0.05177
infant_mortality	-0.201128	0.029567	-6.803	7.16e-10 ***
mmr	-0.005843	0.002139	-2.731	0.00744 **
min_wage	2.142917	0.469984	4.560	1.43e-05 ***
oop_health_exp	0.005479	0.012767	0.429	0.66871
phys_per_1k	-0.132905	0.213889	-0.621	0.53574
labor_force_pct	-0.011836	0.021204	-0.558	0.57792
tax_rev_pct	0.026486	0.037235	0.711	0.47852
total_tax_rate	0.029298	0.014252	2.056	0.04237 *

*** Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.929 on 102 degrees of freedom
Multiple R-squared: 0.9421, Adjusted R-squared: 0.9313
F-statistic: 87.37 on 19 and 102 DF, p-value: < 2.2e-16

{r}

vif(full_model)

density_pop	agri_land_pct	birth_rate	co2	cpi	cpi_pct_change	fertility_rate	gas_price
1.747831	1.368671	63.461464	27.010171	2.937617	3.171808	50.363526	2.665979
gdp	primary_enroll_pct	tertiary_enroll_pct	infant_mortality	mmr	min_wage	oop_health_exp	phys_per_1k
26.105267	1.444301	4.873012	10.758835	6.709072	3.523855	1.811784	3.740211
labor_force_pct	tax_rev_pct	total_tax_rate					
1.424534	1.910437	1.387837					

Multicollinearity Checks (4)

eigprop(1)

```
{r}
eigprop(full_model, Inter = TRUE, prop = 0.5)
```

call:
eigprop(mod = full_model, Inter = TRUE, prop = 0.5)

	Eigenvalues	CI (Intercept)	density_pop	agri_land_pct	birth_rate	co2	cpi	cpi_pct_change	fertility_rate	gas_price	gdp	primary_enroll_pct
tertiary_enroll_pct												
1	16.3148	1.0000	0.0001	0.0004	0.0000	0.0000	0.0000	0.0002	0.0000	0.0001	0.0000	0.0000
2	2.0930	2.7919	0.0000	0.0000	0.0000	0.0001	0.0000	0.0000	0.0004	0.0001	0.0001	0.0000
3	0.3341	6.9885	0.0000	0.0005	0.0270	0.0000	0.0000	0.0000	0.0151	0.0002	0.0056	0.0000
4	0.2350	8.3320	0.0000	0.0132	0.0108	0.0003	0.0000	0.0000	0.0707	0.0002	0.0015	0.0000
5	0.1966	9.1098	0.0000	0.0024	0.0253	0.0001	0.0000	0.0001	0.1652	0.0000	0.0001	0.0000
6	0.1740	9.6818	0.0000	0.0012	0.5344	0.0000	0.0001	0.0000	0.0020	0.0000	0.0003	0.0000
7	0.1422	10.7127	0.0000	0.0065	0.1003	0.0000	0.0011	0.0000	0.0270	0.0000	0.0108	0.0000
8	0.1219	11.5688	0.0000	0.0440	0.0039	0.0035	0.0002	0.0000	0.0530	0.0049	0.0193	0.0000
9	0.0965	13.0050	0.0000	0.0042	0.0020	0.0003	0.0001	0.0001	0.0308	0.0004	0.0010	0.0000
10	0.0749	14.7571	0.0000	0.0144	0.0101	0.0004	0.0005	0.0001	0.0041	0.0014	0.0294	0.0000
11	0.0685	15.4289	0.0001	0.1103	0.0702	0.0027	0.0008	0.0003	0.0041	0.0037	0.0003	0.0001
12	0.0503	18.0041	0.0000	0.0126	0.0019	0.0003	0.0009	0.0001	0.0000	0.0016	0.3236	0.0000
13	0.0348	21.6615	0.0000	0.0000	0.0256	0.0029	0.0003	0.0003	0.0163	0.0055	0.0099	0.0000
14	0.0293	23.5873	0.0001	0.4442	0.0873	0.0021	0.0009	0.0000	0.0189	0.0095	0.0792	0.0000
15	0.0159	31.9872	0.0001	0.2659	0.0056	0.0024	0.0431	0.0005	0.0205	0.0000	0.2203	0.0013
16	0.0107	39.0010	0.0013	0.0206	0.0045	0.0009	0.0079	0.0090	0.0064	0.0026	0.0010	0.0000

eigprop(2)

16	0.0107	39.0010	0.0013	0.0206	0.0045	0.0009	0.0079	0.0090	0.0064	0.0026	0.0010	0.0000	0.3870
17	0.0045	59.9768	0.0270	0.0374	0.0476	0.0123	0.0391	0.1167	0.1320	0.0022	0.0237	0.0012	0.4341
18	0.0017	98.7012	0.0000	0.0188	0.0298	0.9640	0.0041	0.0011	0.0357	0.9534	0.0465	0.0001	0.1321
19	0.0010	125.0148	0.1386	0.0008	0.0007	0.0026	0.0478	0.7458	0.3877	0.0059	0.0296	0.0445	0.0002
20	0.0002	310.7134	0.8327	0.0027	0.0127	0.0051	0.8531	0.1258	0.0100	0.0082	0.1975	0.9527	0.0001
1	0.0001	0.0001	0.0002	0.0004	0.0002	0.0001	0.0002	0.0002	0.0003				
2	0.0037	0.0115	0.0057	0.0005	0.0082	0.0000	0.0007	0.0000	0.0000				
3	0.0019	0.0274	0.0968	0.1263	0.0025	0.0002	0.0046	0.0008	0.0008				
4	0.0011	0.0790	0.0044	0.0116	0.0571	0.0011	0.0342	0.0053	0.0053				
5	0.0026	0.0862	0.0007	0.0820	0.0674	0.0004	0.0012	0.0120	0.0120				
6	0.0018	0.0000	0.0074	0.1135	0.0011	0.0001	0.0264	0.0226	0.0226				
7	0.0031	0.0000	0.1023	0.0020	0.1662	0.0001	0.1472	0.0158	0.0158				
8	0.0000	0.0358	0.0651	0.0877	0.0548	0.0035	0.0253	0.0000	0.0000				
9	0.0039	0.0112	0.0880	0.0190	0.0669	0.0015	0.0745	0.4971	0.4971				
10	0.0050	0.0193	0.0119	0.0007	0.2696	0.0046	0.1019	0.2125	0.2125				
11	0.0032	0.0693	0.2337	0.2711	0.0341	0.0237	0.0611	0.0002	0.0002				
12	0.0048	0.0092	0.1183	0.0004	0.0819	0.0000	0.1652	0.0506	0.0506				
13	0.8343	0.4411	0.0002	0.0444	0.0079	0.0050	0.0123	0.0086	0.0086				
14	0.0064	0.0054	0.0136	0.1385	0.0071	0.1682	0.0530	0.0018	0.0018				
15	0.0181	0.0624	0.1457	0.0034	0.0014	0.1549	0.0360	0.0433	0.0433				
16	0.0262	0.0506	0.0003	0.0026	0.0000	0.5161	0.1201	0.0422	0.0422				
17	0.0661	0.0311	0.0027	0.0375	0.0325	0.0153	0.0821	0.0472	0.0472				
18	0.0034	0.0353	0.0001	0.0469	0.1237	0.0084	0.0045	0.0000	0.0000				
19	0.0022	0.0014	0.0461	0.0094	0.0031	0.0678	0.0079	0.0294	0.0294				
20	0.0121	0.0235	0.0568	0.0019	0.0144	0.0292	0.0416	0.0104	0.0104				

```
=====
Row 6==> agri_land_pct, proportion 0.534426 >= 0.50
Row 18==> birth_rate, proportion 0.964002 >= 0.50
Row 20==> co2, proportion 0.853100 >= 0.50
Row 19==> cpi, proportion 0.745796 >= 0.50
Row 18==> fertility_rate, proportion 0.953441 >= 0.50
Row 20==> gdp, proportion 0.952748 >= 0.50
Row 13==> infant_mortality, proportion 0.834326 >= 0.50
Row 16==> labor_force_pct, proportion 0.516077 >= 0.50
```

eigprop(3)

```
{r}
reduced_model_1 <- lm(life_expectancy ~ . - gdp, data = world_model_data)
vif(reduced_model_1)
eigprop(reduced_model_1, prop = 0.5)
```

	density_pop	agri_land_pct	birth_rate	co2	cpi	cpi_pct_change	fertility_rate	gas_price
1	1.736214	1.351445	63.175999	2.804664	2.846128	3.170967	49.866780	2.091188
2	primary_enroll_pct	tertiary_enroll_pct	infant_mortality	mmr	min_wage	oop_health_exp	phys_per_1k	labor_force_pct
3	1.442346	4.870396	10.647647	6.578803	3.225270	1.811530	3.705269	1.345647
4	tax_rev_pct	total_tax_rate						
5	1.851426	1.361658						

Call:
eigprop(mod = reduced_model_1, prop = 0.5)

	Eigenvalues	CI (Intercept)	density_pop	agri_land_pct	birth_rate	co2	cpi	cpi_pct_change	fertility_rate	gas_price	primary_enroll_pct
1	15.3301	1.0000	0.0000	0.0002	0.0005	0.0000	0.0001	0.0000	0.0003	0.0000	0.0000
2	2.0872	2.7101	0.0000	0.0000	0.0000	0.0001	0.0001	0.0000	0.0004	0.0001	0.0001
3	0.3340	6.7747	0.0000	0.0005	0.0277	0.0000	0.0003	0.0000	0.0153	0.0002	0.0071
4	0.2338	8.0979	0.0001	0.0141	0.0089	0.0003	0.0000	0.0000	0.0723	0.0002	0.0024
5	0.1961	8.8413	0.0000	0.0020	0.0221	0.0002	0.0003	0.0001	0.1653	0.0000	0.0004
6	0.1735	9.3999	0.0000	0.0012	0.5600	0.0000	0.0008	0.0000	0.0009	0.0000	0.0001
7	0.1399	10.4675	0.0000	0.0093	0.0865	0.0000	0.0106	0.0000	0.0221	0.0000	0.0104
8	0.1212	11.2473	0.0001	0.0416	0.0063	0.0037	0.0023	0.0000	0.0586	0.0052	0.0255
9	0.0962	12.6250	0.0000	0.0039	0.0012	0.0004	0.0010	0.0001	0.0287	0.0006	0.0014
10	0.0746	14.3377	0.0001	0.0216	0.0051	0.0002	0.0042	0.0001	0.0048	0.0009	0.0364
11	0.0670	15.1263	0.0004	0.1160	0.0744	0.0025	0.0111	0.0005	0.0035	0.0036	0.0010
12	0.0501	17.4909	0.0001	0.0086	0.0029	0.0003	0.0106	0.0001	0.0000	0.0017	0.4262
13	0.2334										0.0033

eigprop(4)

13	0.0348	20.9978	0.0001	0.0001	0.0260	0.0029	0.0031	0.0003	0.0163	0.0056	0.0130	0.0020
14	0.0293	22.8666	0.0002	0.4419	0.0881	0.0021	0.0082	0.0000	0.0188	0.0096	0.0985	0.0199
15	0.0146	32.4322	0.0006	0.2562	0.0078	0.0022	0.5980	0.0013	0.0246	0.0000	0.2953	0.0075
16	0.0107	37.8102	0.0037	0.0190	0.0044	0.0010	0.0866	0.0091	0.0060	0.0027	0.0008	0.3868
17	0.0044	58.8570	0.0845	0.0430	0.0476	0.0130	0.2292	0.1424	0.1558	0.0020	0.0184	0.4396
18	0.0017	95.6909	0.0001	0.0191	0.0305	0.9709	0.0309	0.0005	0.0321	0.9655	0.0567	0.1312
19	0.0008	134.6223	0.9101	0.0018	0.0000	0.0002	0.0026	0.8454	0.3742	0.0021	0.0062	0.0013
1	infant_mortality	mmr	min_wage	oop_health_exp	phys_per_1k	labor_force_pct	tax_rev_pct	total_tax_rate				
2	0.0001	0.0002	0.0003	0.0004	0.0003	0.0001	0.0003	0.0003				
3	0.0037	0.0117	0.0063	0.0005	0.0084	0.0000	0.0008	0.0000				
4	0.0019	0.0278	0.1055	0.1266	0.0026	0.0002	0.0047	0.0009				
5	0.0010	0.0785	0.0050	0.0133	0.0530	0.0012	0.0394	0.0055				
6	0.0026	0.0922	0.0005	0.0824	0.0680	0.0005	0.0004	0.0117				
7	0.0019	0.0002	0.0073	0.1140	0.0012	0.0001	0.0218	0.0240				
8	0.0032	0.0000	0.1334	0.0071	0.1891	0.0001	0.1435	0.0235				
9	0.0000	0.0412	0.0554	0.0790	0.0404	0.0037	0.0339	0.0012				
10	0.0039	0.0151	0.1069	0.0217	0.0662	0.0016	0.0838	0.4895				
11	0.0059	0.0107	0.0028	0.0069	0.2935	0.0035	0.1111	0.2192				
12	0.0023	0.0667	0.2503	0.2603	0.0137	0.0326	0.0643	0.0054				
13	0.0050	0.0100	0.1244	0.0012	0.0812	0.0003	0.1604	0.0492				
14	0.8427	0.4497	0.0002	0.0442	0.0079	0.0054	0.0127	0.0088				
15	0.0065	0.0054	0.0155	0.1381	0.0072	0.1810	0.0553	0.0019				
16	0.0225	0.0730	0.1505	0.0036	0.0018	0.1483	0.0359	0.0468				
17	0.0256	0.0497	0.0005	0.0026	0.0000	0.5411	0.1260	0.0417				
18	0.0636	0.0274	0.0032	0.0382	0.0323	0.0145	0.0821	0.0490				
19	0.0035	0.0359	0.0001	0.0472	0.1249	0.0096	0.0046	0.0000				
20	0.0040	0.0049	0.0318	0.0129	0.0083	0.0562	0.0190	0.0216				

=====
Row 6==> agri_land_pct, proportion 0.559976 >= 0.50
Row 18==> birth_rate, proportion 0.970918 >= 0.50
Row 15==> co2, proportion 0.597967 >= 0.50
Row 19==> cpi, proportion 0.845373 >= 0.50
Row 18==> fertility_rate, proportion 0.965481 >= 0.50
Row 13==> infant_mortality, proportion 0.842717 >= 0.50
Row 16==> labor_force_pct, proportion 0.541070 >= 0.50

eigprop(5)

```
[r]
# Continue removing variables with high VIF and variance decomposition proportions
reduced_model_2 <- lm(life_expectancy ~ . - gdp - birth_rate, data = world_model_data)
vif(reduced_model_2)

eigprop(reduced_model_2, prop = 0.5)
```

	density_pop	agri_land_pct	co2	cpi	cpi_pct_change	fertility_rate	gas_price	primary_enroll_pct
	1.676460	1.294788	2.715912	2.842742	3.130331	5.094977	1.998508	1.336820
tertiary_enroll_pct		infant_mortality	mmr	min_wage	oop_health_exp	phys_per_1k	labor_force_pct	tax_rev_pct
	4.733005	10.466364	6.324083	3.223383	1.745877	3.192812	1.328959	1.846830
total_tax_rate	1.359462							

```
Call:
eigprop(mod = reduced_model_2, prop = 0.5)
```

	Eigenvalues	CI (Intercept)	density_pop	agri_land_pct	co2	cpi	cpi_pct_change	fertility_rate	gas_price	primary_enroll_pct	tertiary_enroll_pct
infant_mortality											
1	14.5123	1.0000	0.0000	0.0002	0.0006	0.0001	0.0000	0.0003	0.0001	0.0002	0.0002
0.0001											
2	1.9454	2.7313	0.0000	0.0000	0.0000	0.0001	0.0000	0.0007	0.0013	0.0001	0.0049
0.0047											
3	0.3324	6.6077	0.0000	0.0004	0.0274	0.0004	0.0000	0.0139	0.0016	0.0076	0.0000
0.0025											
4	0.2289	7.9631	0.0001	0.0190	0.0074	0.0000	0.0000	0.0875	0.0016	0.0036	0.0010
0.0003											
5	0.1941	8.6461	0.0000	0.0004	0.0246	0.0003	0.0001	0.1565	0.0003	0.0012	0.0005
0.0021											
6	0.1735	9.1465	0.0000	0.0010	0.5889	0.0008	0.0000	0.0007	0.0002	0.0001	0.0000
0.0019											
7	0.1398	10.1871	0.0000	0.0079	0.0940	0.0112	0.0000	0.0255	0.0001	0.0125	0.0000
0.0034											
8	0.1049	11.7645	0.0001	0.0392	0.0005	0.0032	0.0001	0.0870	0.0415	0.0199	0.0025
0.0005											
9	0.0933	12.4737	0.0002	0.0000	0.0035	0.0058	0.0004	0.0027	0.0297	0.0131	0.0009
0.0097											
10	0.0742	13.9836	0.0000	0.0332	0.0011	0.0018	0.0000	0.0080	0.0084	0.0308	0.0001
0.0044											
11	0.0606	15.4724	0.0001	0.1848	0.0979	0.0055	0.0002	0.0002	0.1085	0.0010	0.0024
0.0288											
12	0.0496	17.1020	0.0001	0.0001	0.0128	0.0119	0.0002	0.0006	0.0400	0.4625	0.0034
0.0000											

eigprop(6)

13	0.0331	20.9535	0.0003	0.0668	0.0008	0.0063	0.0004	0.0046	0.0952	0.0003	0.0001	0.0317
0.7960												
14	0.0279	22.8249	0.0000	0.3193	0.0870	0.0074	0.0000	0.0231	0.3645	0.0929	0.0281	0.0762
0.0321												
15	0.0143	31.8511	0.0005	0.2693	0.0093	0.6271	0.0010	0.0277	0.0325	0.3373	0.0040	0.2117
0.0182												
16	0.0107	36.8960	0.0042	0.0162	0.0034	0.1146	0.0099	0.0050	0.0019	0.0000	0.4186	0.0118
0.0263												
17	0.0043	57.9023	0.0830	0.0400	0.0408	0.2008	0.1416	0.1744	0.2633	0.0098	0.5373	0.0002
0.0649												
18	0.0008	130.9749	0.9113	0.0021	0.0001	0.0030	0.8461	0.3817	0.0092	0.0070	0.0011	0.0001
0.0039												

	mmr	min_wage	oop_health_exp	phys_per_1k	labor_force_pct	tax_rev_pct	total_tax_rate
1	0.0002	0.0004	0.0005	0.0004	0.0001	0.0003	0.0004
2	0.0149	0.0067	0.0008	0.0104	0.0000	0.0007	0.0000
3	0.0355	0.1065	0.1292	0.0030	0.0002	0.0045	0.0007
4	0.0537	0.0036	0.0203	0.0638	0.0012	0.0460	0.0060
5	0.0959	0.0007	0.0767	0.1085	0.0007	0.0001	0.0112
6	0.0002	0.0076	0.1180	0.0008	0.0001	0.0209	0.0245
7	0.0002	0.1284	0.0059	0.2057	0.0001	0.1511	0.0230
8	0.0200	0.1909	0.1676	0.0015	0.0081	0.0013	0.0837
9	0.0335	0.0131	0.0042	0.0866	0.0122	0.1107	0.4333
10	0.0082	0.0008	0.0318	0.3750	0.0004	0.1466	0.1900
11	0.1688	0.2492	0.1780	0.0019	0.0173	0.0299	0.0143
12	0.0417	0.0892	0.0059	0.0861	0.0005	0.1624	0.0422
13	0.3697	0.0035	0.1202	0.0175	0.0359	0.0007	0.0071
14	0.0034	0.0128	0.0713	0.0016	0.1572	0.0636	0.0061
15	0.0850	0.1508	0.0048	0.0043	0.1606	0.0291	0.0485
16	0.0439	0.0013	0.0020	0.0002	0.5336	0.1299	0.0386
17	0.0204	0.0026	0.0504	0.0219	0.0157	0.0833	0.0489
18	0.0046	0.0319	0.0126	0.0108	0.0561	0.0188	0.0215

```
=====> agri_land_pct, proportion 0.588932 >= 0.50
Row 15=> co2, proportion 0.627062 >= 0.50
Row 18=> cpi, proportion 0.846062 >= 0.50
Row 17=> primary_enroll_pct, proportion 0.537295 >= 0.50
Row 13=> infant_mortality, proportion 0.796028 >= 0.50
Row 16=> labor_force_pct, proportion 0.533559 >= 0.50
```

Model after Collinearity Checks

```
{r}
maybe_full_model <- update(reduced_model_2, . ~ .)

summary(maybe_full_model)
corr_matrix_maybe <- cor(world_model_data |> dplyr::select(-c(gdp, birth_rate)))

Call:
lm(formula = life_expectancy ~ density_pop + agri_land_pct +
    co2 + cpi + cpi_pct_change + fertility_rate + gas_price +
    primary_enroll_pct + tertiary_enroll_pct + infant_mortality +
    mmr + min_wage + oop_health_exp + phys_per_1k + labor_force_pct +
    tax_rev_pct + total_tax_rate, data = world_model_data)

Residuals:
    Min       1Q   Median       3Q      Max
-4.9778 -1.1014  0.0888  1.2397  4.0284
```

Residuals:

	Min	1Q	Median	3Q	Max
	-4.9778	-1.1014	0.0888	1.2397	4.0284

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	6.979e+01	4.792e+00	14.564	< 2e-16	***
density_pop	4.985e-01	1.926e-01	2.588	0.01102	*
agri_land_pct	-1.944e-02	9.869e-03	-1.970	0.05153	.
co2	4.046e-02	1.333e-01	0.303	0.76215	
cpi	4.984e-01	8.632e-01	0.577	0.56488	
cpi_pct_change	-4.203e-01	3.976e-01	-1.057	0.29287	
fertility_rate	-3.849e-01	3.082e-01	-1.249	0.21464	
gas_price	3.603e-01	8.149e-01	0.442	0.65935	
primary_enroll_pct	2.744e-03	1.769e-02	0.155	0.87700	
tertiary_enroll_pct	2.396e-02	1.328e-02	1.805	0.07398	.
infant_mortality	-2.045e-01	2.953e-02	-6.925	3.72e-10	***
mmr	-5.533e-03	2.103e-03	-2.631	0.00982	**
min_wage	2.414e+00	4.552e-01	5.302	6.46e-07	***
oop_health_exp	6.902e-03	1.269e-02	0.544	0.58773	
phys_per_1k	-2.314e-01	2.001e-01	-1.156	0.25029	
labor_force_pct	-4.614e-05	2.074e-02	-0.002	0.99823	
tax_rev_pct	1.447e-02	3.707e-02	0.390	0.69718	
total_tax_rate	3.369e-02	1.428e-02	2.358	0.02023	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.954 on 104 degrees of freedom
Multiple R-squared: 0.9395, Adjusted R-squared: 0.9296
F-statistic: 94.96 on 17 and 104 DF, p-value: < 2.2e-16

Model after collinearity checks and removed infant_mortality, mmr, fertility_rate

```
{r}
full_model <- lm(life_expectancy ~ . - gdp - birth_rate - infant_mortality - mmr - fertility_rate, data = world_model_data)
summary(full_model)
vif(full_model)

corr_matrix_final <- cor(world_model_data |> dplyr::select(-c(gdp, birth_rate, infant_mortality, mmr, fertility_rate)))

Call:
lm(formula = life_expectancy ~ . - gdp - birth_rate - infant_mortality - mmr - fertility_rate, data = world_model_data)

Residuals:
    Min       1Q   Median       3Q      Max
-13.604  -1.642   0.265   2.039   7.012

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  57.8723004  8.7716056   6.598 1.64e-09 ***
density_pop   1.1578984  0.3393642   3.412 0.000911 ***
agri_land_pct -0.0489711  0.0179551  -2.727 0.007462 **
co2           0.4246796  0.2454492   1.730 0.086475 .
cpi           0.5686890  1.6329841   0.348 0.728336
cpi_pct_change -0.9697808  0.7326524  -1.324 0.188438
gas_price    -1.7624887  1.5219651  -1.158 0.249428
primary_enroll_pct 0.0220877  0.0319215   0.692 0.490475
tertiary_enroll_pct 0.0835546  0.0237115   3.524 0.000627 ***
min_wage     2.7207810  0.8574076   3.173 0.001968 **
oop_health_exp -0.0342914  0.0234145  -1.465 0.145979
phys_per_1k   0.8190886  0.3614946   2.266 0.025474 *
labor_force_pct -0.0502011  0.0387896  -1.294 0.198387
tax_rev_pct   0.1443180  0.0677674   2.130 0.035497 *
total_tax_rate -0.0003174  0.0263185  -0.012 0.990400
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.73 on 107 degrees of freedom
Multiple R-squared:  0.773,    Adjusted R-squared:  0.7433
F-statistic: 26.03 on 14 and 107 DF,  p-value: < 2.2e-16
```

density_pop	agri_land_pct	co2	cpi	cpi_pct_change	gas_price	primary_enroll_pct	tertiary_enroll_pct
1.428208	1.175692	2.525009	2.791057	2.916044	1.912216	1.194392	4.141689
min_wage	oop_health_exp	phys_per_1k	labor_force_pct	tax_rev_pct	total_tax_rate		
3.137285	1.630186	2.857931	1.275238	1.692759	1.265961		

Both direction step using step()

```
{r}
# Step on both directions
step_model <- step(full_model, direction = "both", trace = FALSE)
step_model
step_model$anova$AIC

Call:
lm(formula = life_expectancy ~ density_pop + agri_land_pct +
    co2 + cpi_pct_change + tertiary_enroll_pct + min_wage + phys_per_1k +
    tax_rev_pct, data = world_model_data)

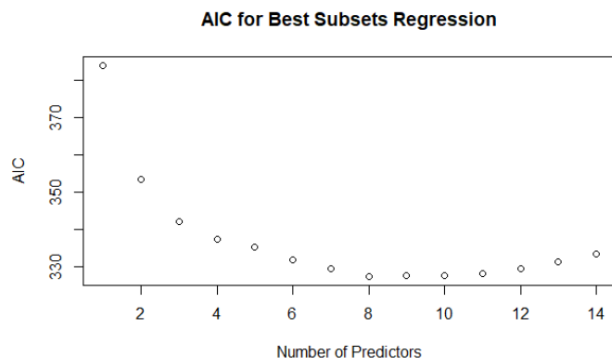
Coefficients:
(Intercept)      density_pop      agri_land_pct          co2      cpi_pct_change  tertiary_enroll_pct      min_wage
phys_per_1k
0.77764      56.15576      0.96727      -0.04646      0.56744      -0.93785      0.07801      2.78394
tax_rev_pct
0.16283

[1] 335.2186 333.2188 331.3620 330.0036 329.6166 329.5950 329.2961
```

AIC Code

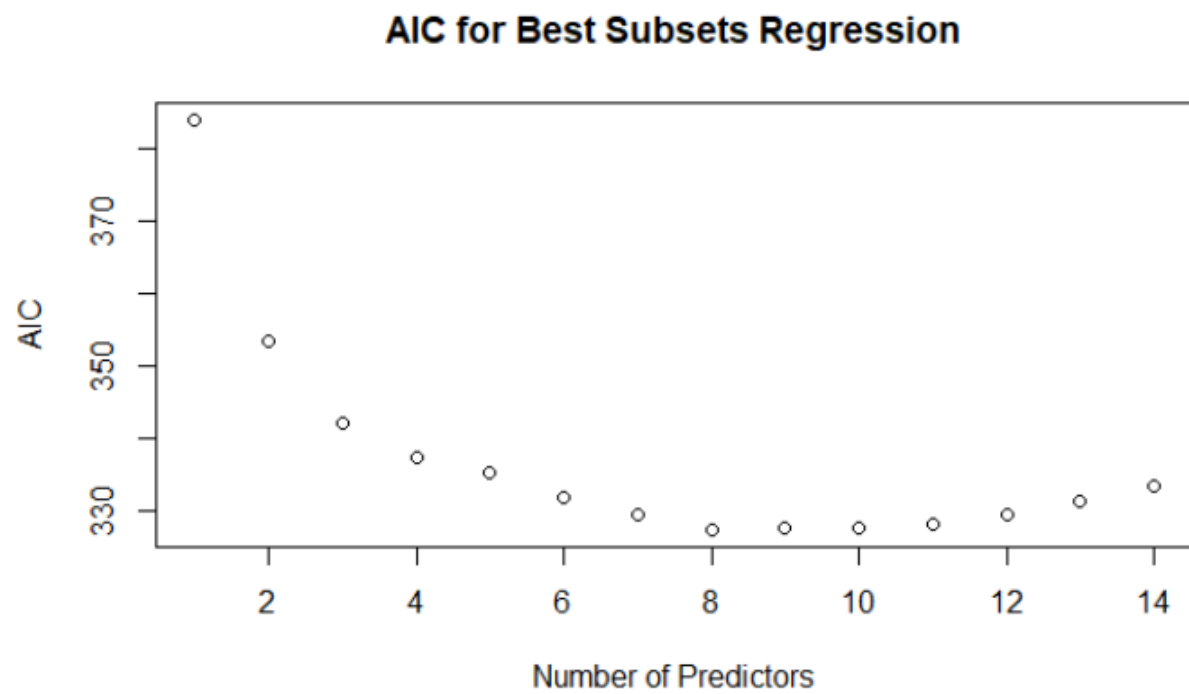
```
{r}
best_subsets <- regsubsets(life_expectancy ~ . - gdp - birth_rate - infant_mortality - mmr - fertility_rate, data = world_model_data, nvmax = 14)
best_summary <- summary(best_subsets)

n <- nrow(world_model_data)
rss <- best_summary$rss
AIC <- n * log(rss/n) + 2 * (1:14)
AIC
plot(1:14, AIC, xlab = "Number of Predictors", ylab = "AIC", main = "AIC for Best Subsets Regression")
```



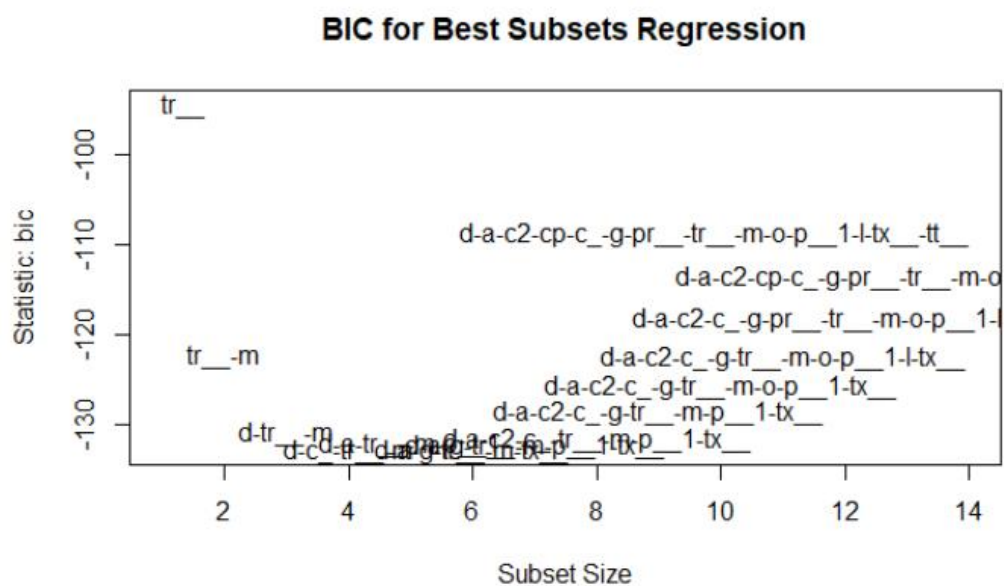
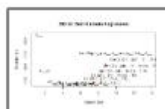
```
[1] 383.9816 353.3029 342.0781 337.3280 335.1972 331.6518 329.4183 327.2961 327.5950
327.6166 328.0036 329.3620 331.2188 333.2186
```

AIC Graph



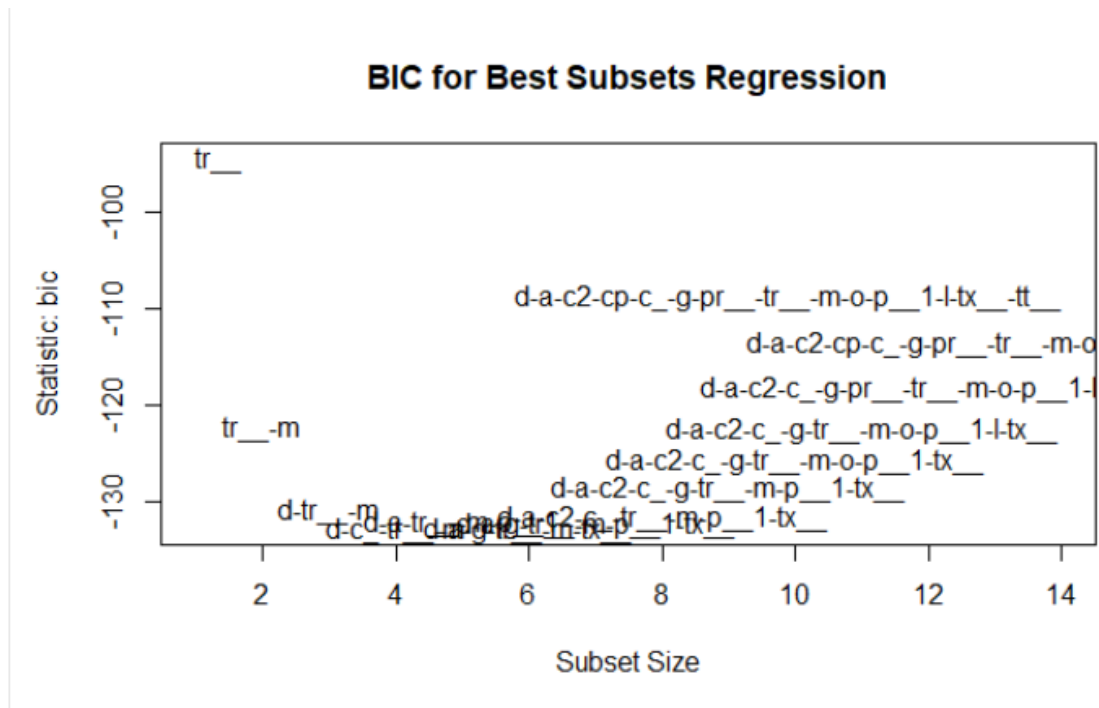
BIC Code

```
{r}
best_summary$bic
subsets(best_subsets, statistic = "bic", main = "BIC for Best Subsets Regression", legend = TRUE)
```



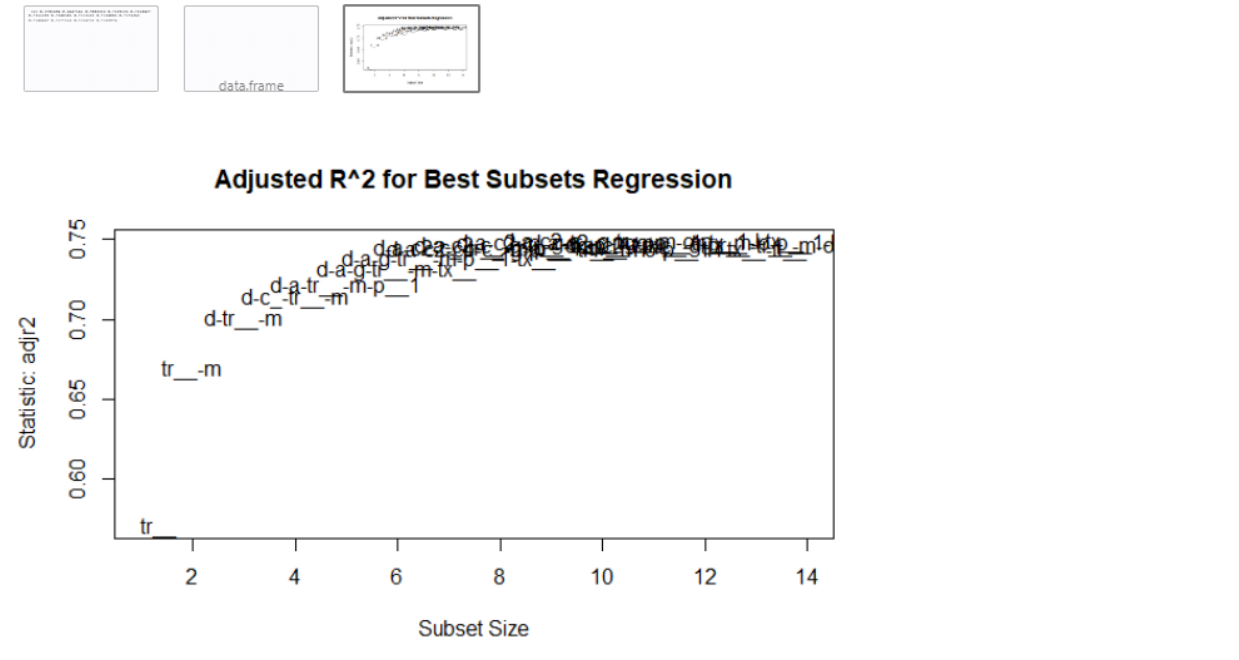
```
[1] -94.53053 -122.40515 -130.82599 -132.77207 -132.09879 -132.84017 -132.26968  
-131.58781 -128.48497 -125.65932 -122.46827 -118.30584 -113.64510 -108.84124
```

BIC Graph



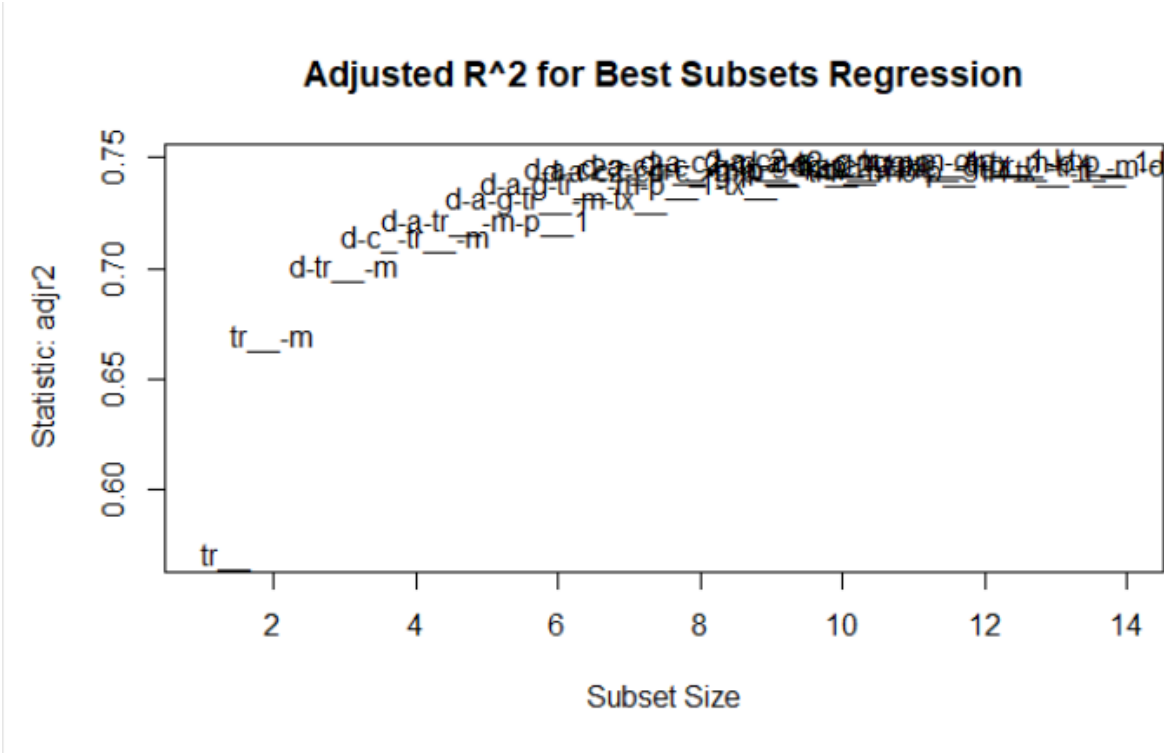
Adjusted R^2 Code

```
{r}
best_summary$adjr2
subsets(best_subsets, statistic = "a", main = "Adjusted R^2 for Best Subsets Regression", legend = TRUE)
```



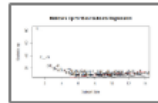
[1] 0.5705698 0.6687181 0.7002322 0.7139433 0.7210827 0.7311592 0.7380505 0.7445123
0.7458005 0.7476362 0.7486867 0.7477113 0.7456742 0.7432976

Adjusted R^2 Graph

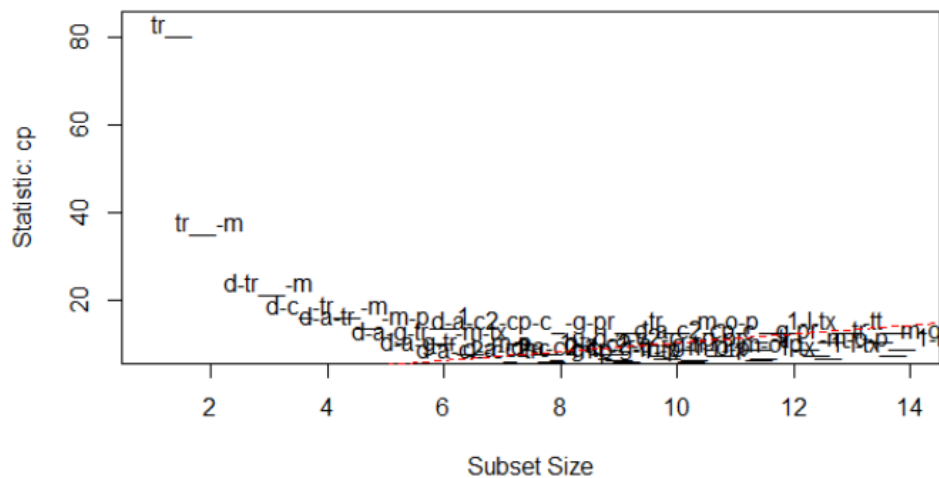


Mallow's CP Code

```
{r}  
best_summary$cp  
  
subsets(best_subsets, statistic = "cp", main = "Mallow's Cp for Best Subsets Regression", legend = TRUE)  
abline(0,1,col="red",lty=2)
```

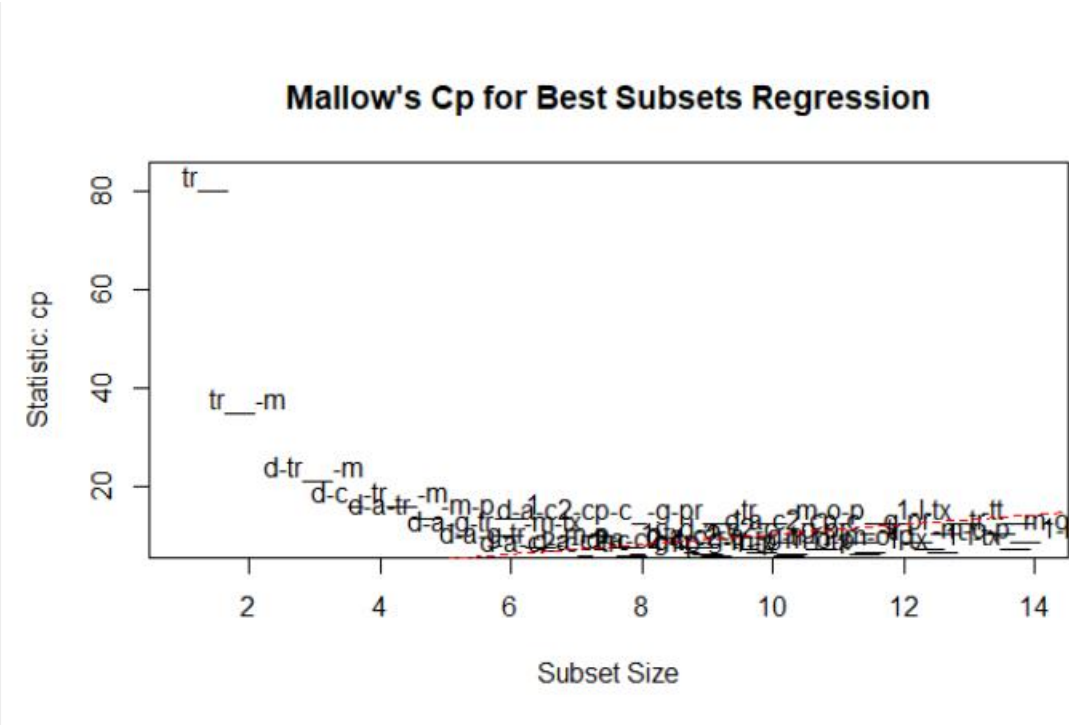


Mallow's Cp for Best Subsets Regression



```
[1] 82.744634 37.572984 23.796154 18.379151 16.038592 12.437875 10.330233 8.465320  
8.907974 9.123974 9.690736 11.125878 13.000145 15.000000
```

Mallow's Cp Graph



Cross Validation

```
{r}
set.seed(123)

train_control <- traincontrol(method = "cv", number = 10)

model_cv5 <- train(life_expectancy ~ density_pop + agri_land_pct +
  tertiary_enroll_pct + min_wage +
  phys_per_1k,
  data = world_model_data,
  method = "lm",
  trcontrol = train_control)
model_cv5$results$RMSE

model_cv9 <- train(life_expectancy ~ density_pop + agri_land_pct +
  co2 + cpi_pct_change + tertiary_enroll_pct + min_wage +
  oop_health_exp + phys_per_1k + tax_rev_pct,
  data = world_model_data,
  method = "lm",
  trcontrol = train_control)

model_cv9$results$RMSE

[1] 3.774645
[1] 3.83021
```

Final model

```
{r}
# Final model summary
summary(best_model)
```

Call:

```
lm(formula = life_expectancy ~ density_pop + agri_land_pct +
  tertiary_enroll_pct + min_wage + phys_per_1k, data = world_model_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-11.8857	-1.9080	0.2675	2.6350	8.1961

Coefficients:

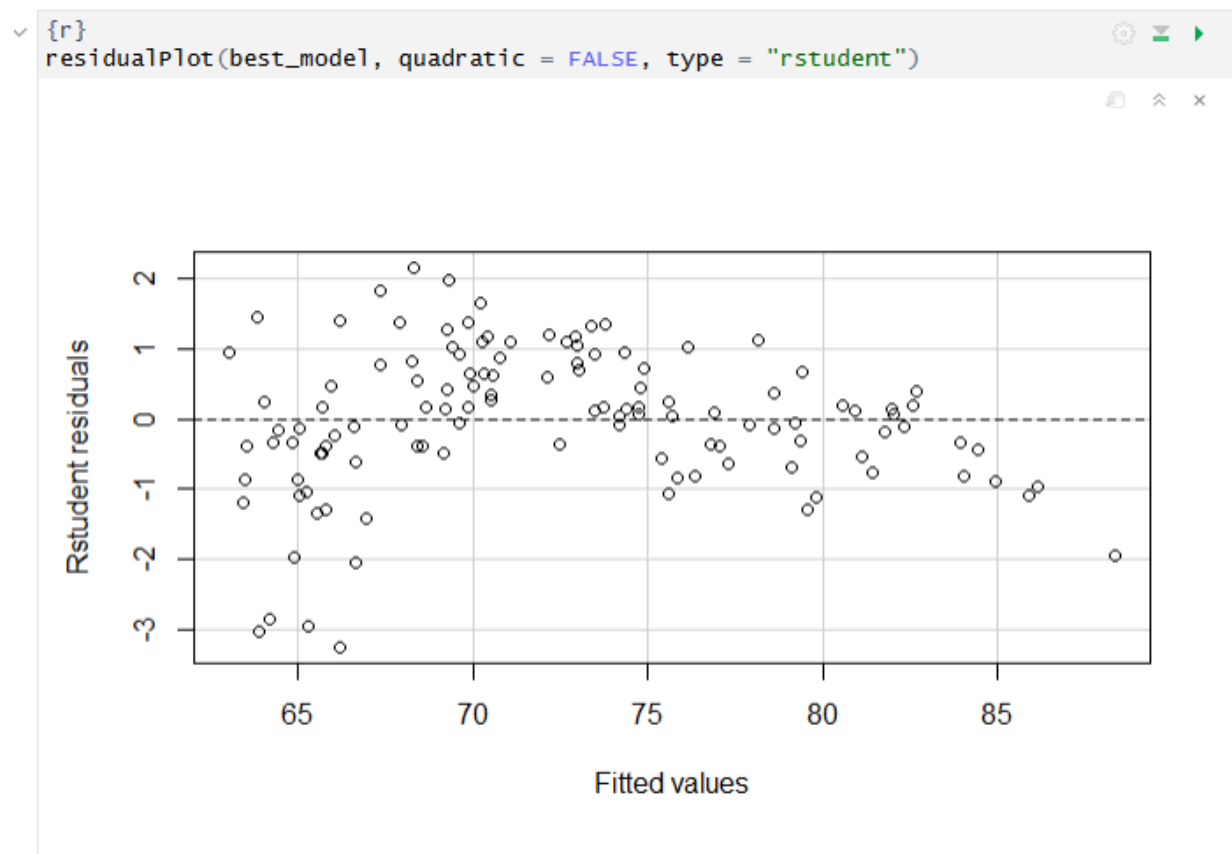
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	60.02400	1.51795	39.543	< 2e-16 ***
density_pop	1.26098	0.30742	4.102	7.64e-05 ***
agri_land_pct	-0.04602	0.01804	-2.551	0.0120 *
tertiary_enroll_pct	0.10705	0.02162	4.951	2.53e-06 ***
min_wage	3.46423	0.69240	5.003	2.03e-06 ***
phys_per_1k	0.81251	0.36391	2.233	0.0275 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.888 on 116 degrees of freedom
Multiple R-squared: 0.7326, Adjusted R-squared: 0.7211
F-statistic: 63.56 on 5 and 116 DF, p-value: < 2.2e-16

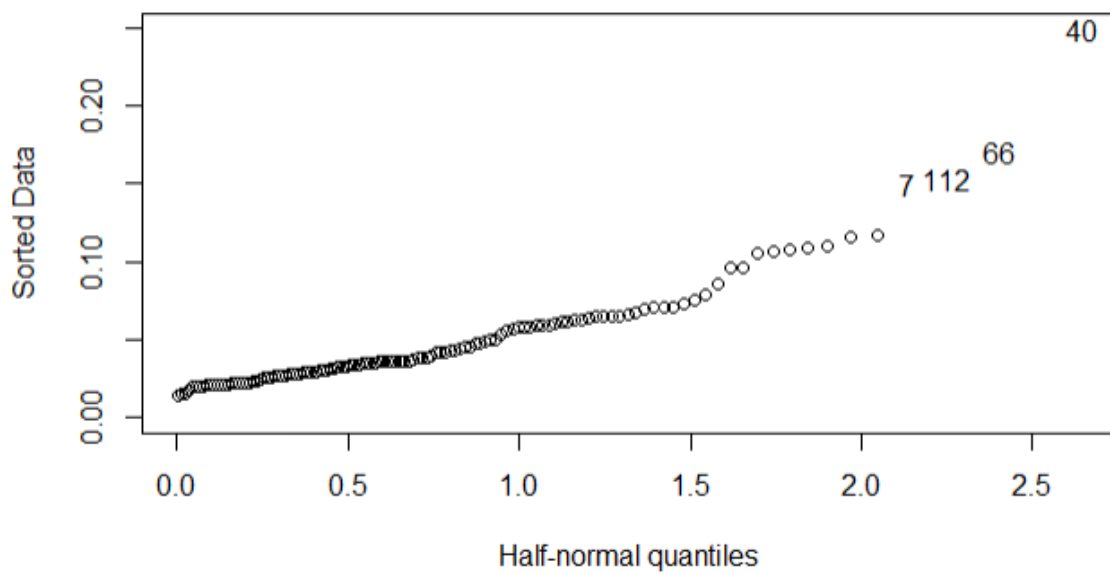
Influential Observations

Studentized Residual plot



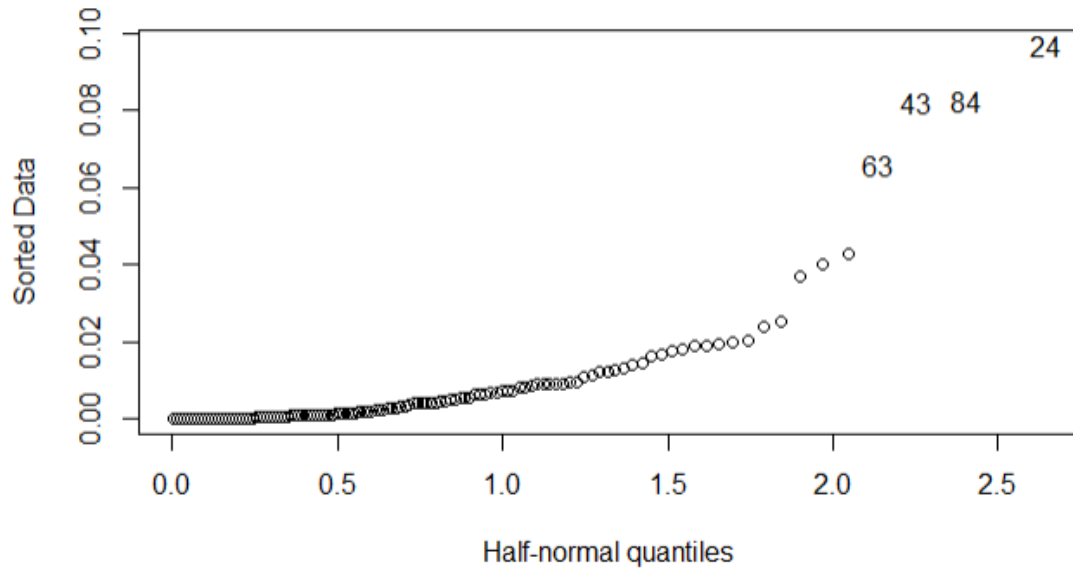
Half-normal hat value plot

```
{r}  
# Compute the hat values from the model  
h <- hatvalues(best_model)  
  
# Use half-normal plot to check for influential observations  
halfnorm(h, nlab = 4)
```

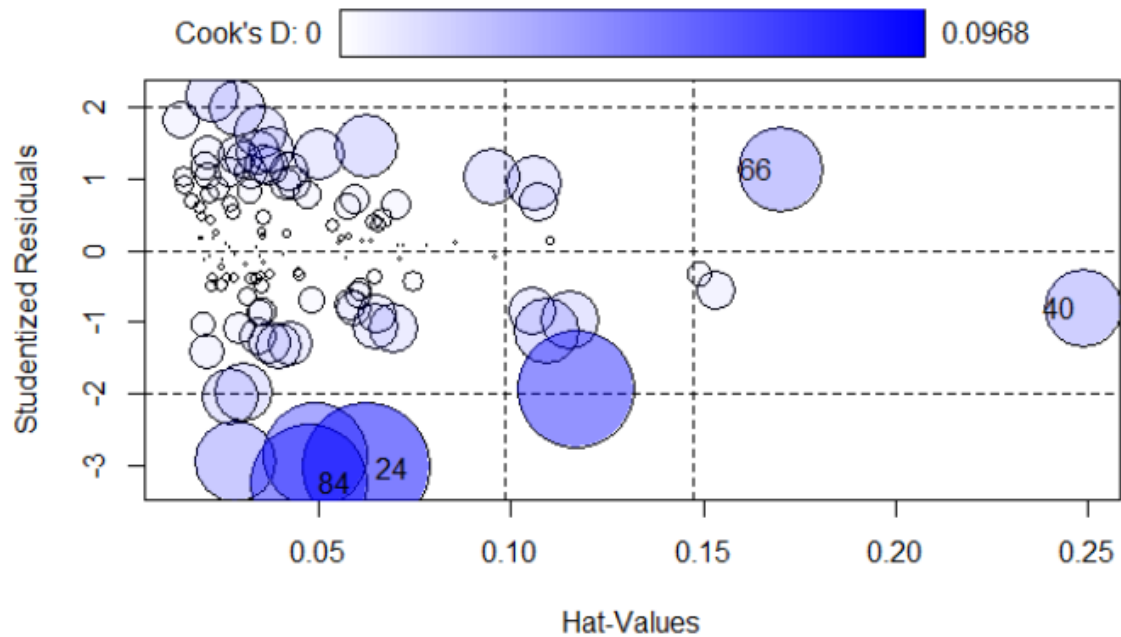
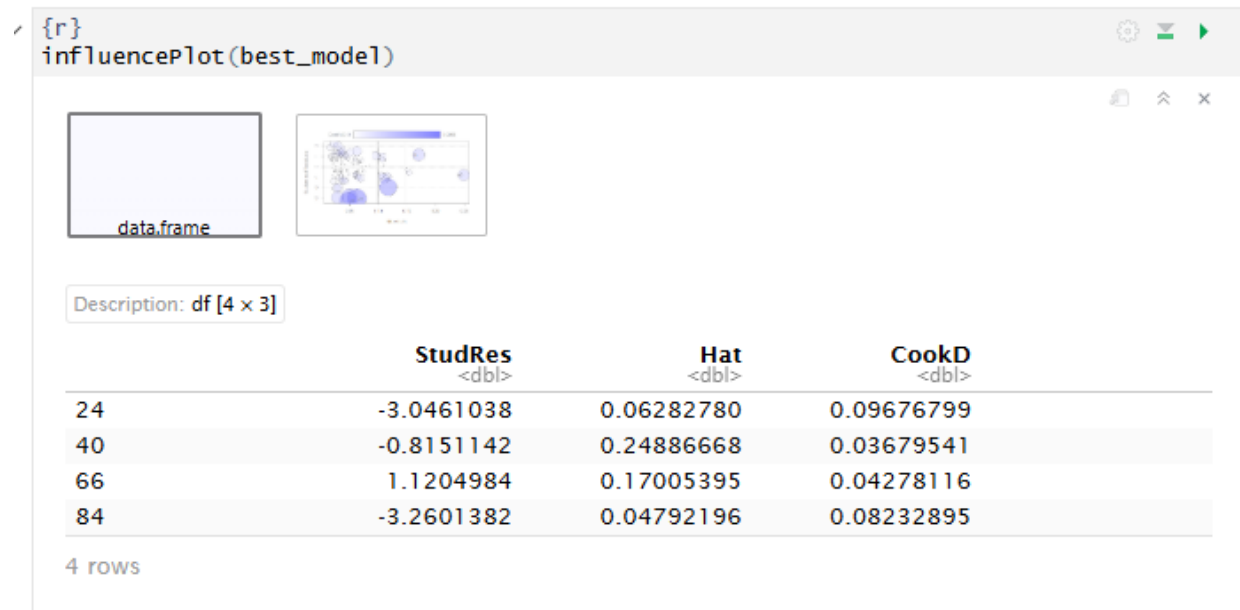


Half-normal Cook's distance plot

```
{r}  
cook <- cooks.distance(best_model)  
  
halfnorm(cook, n = 4)
```

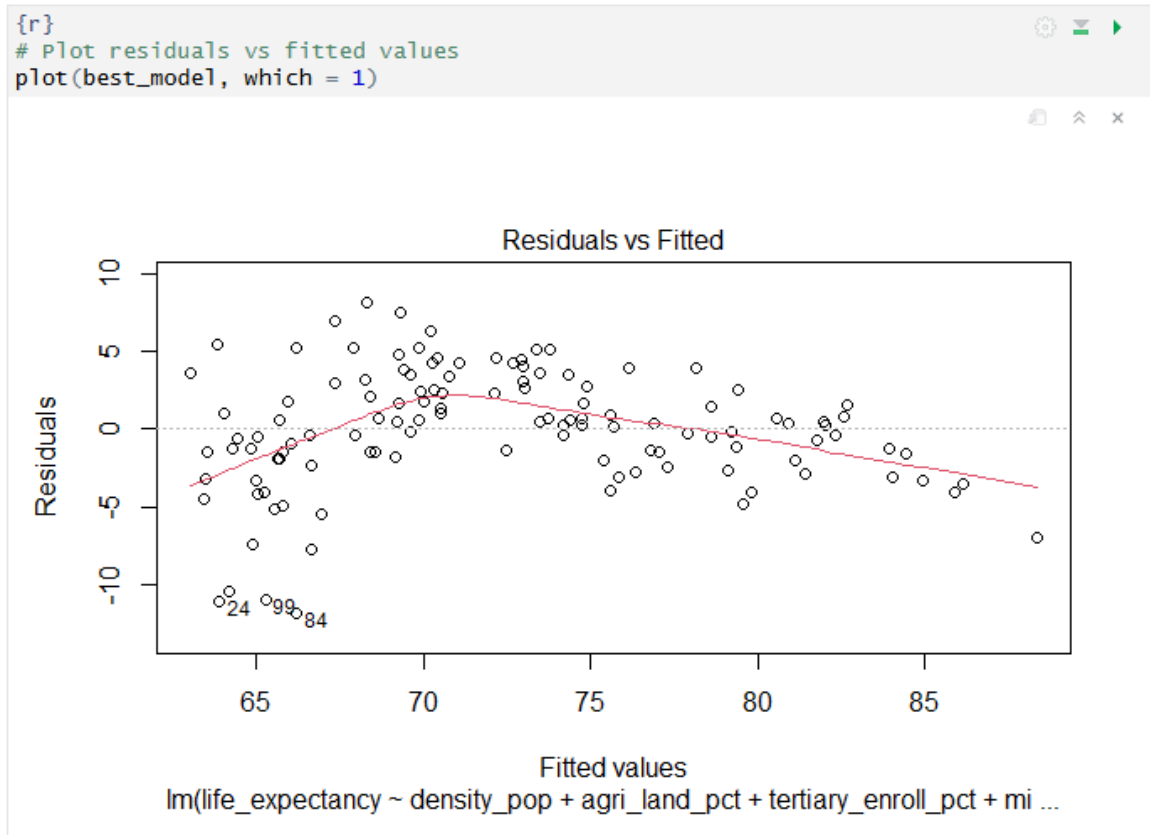


Influence plot

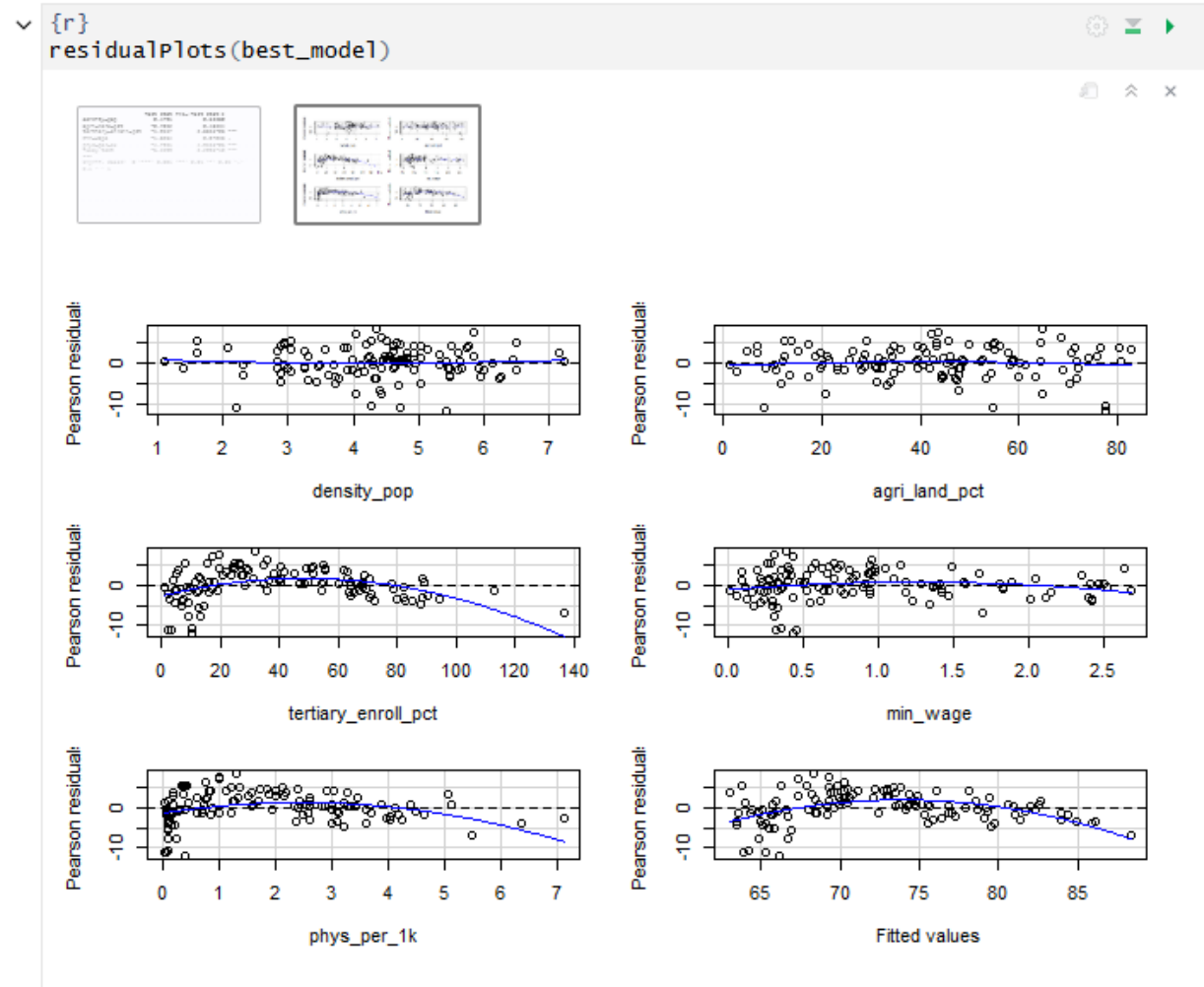


Checking Error Assumptions

Residuals vs fitted plot



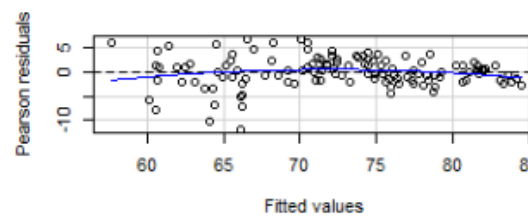
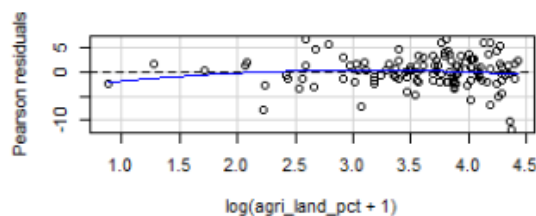
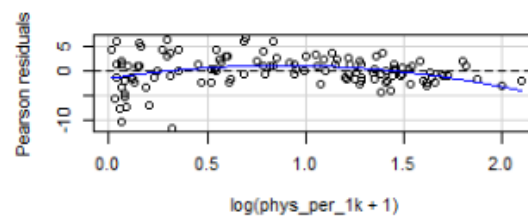
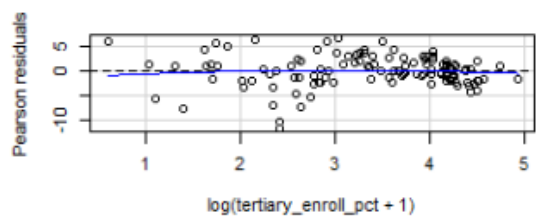
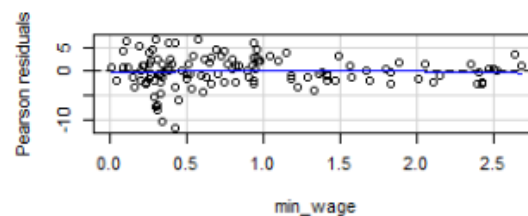
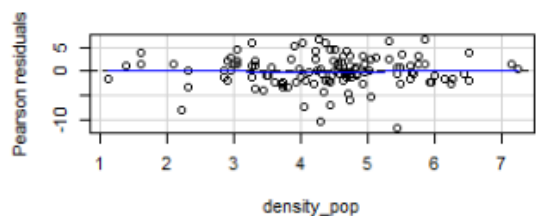
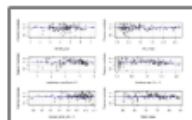
Predictors vs. fitted before transformation



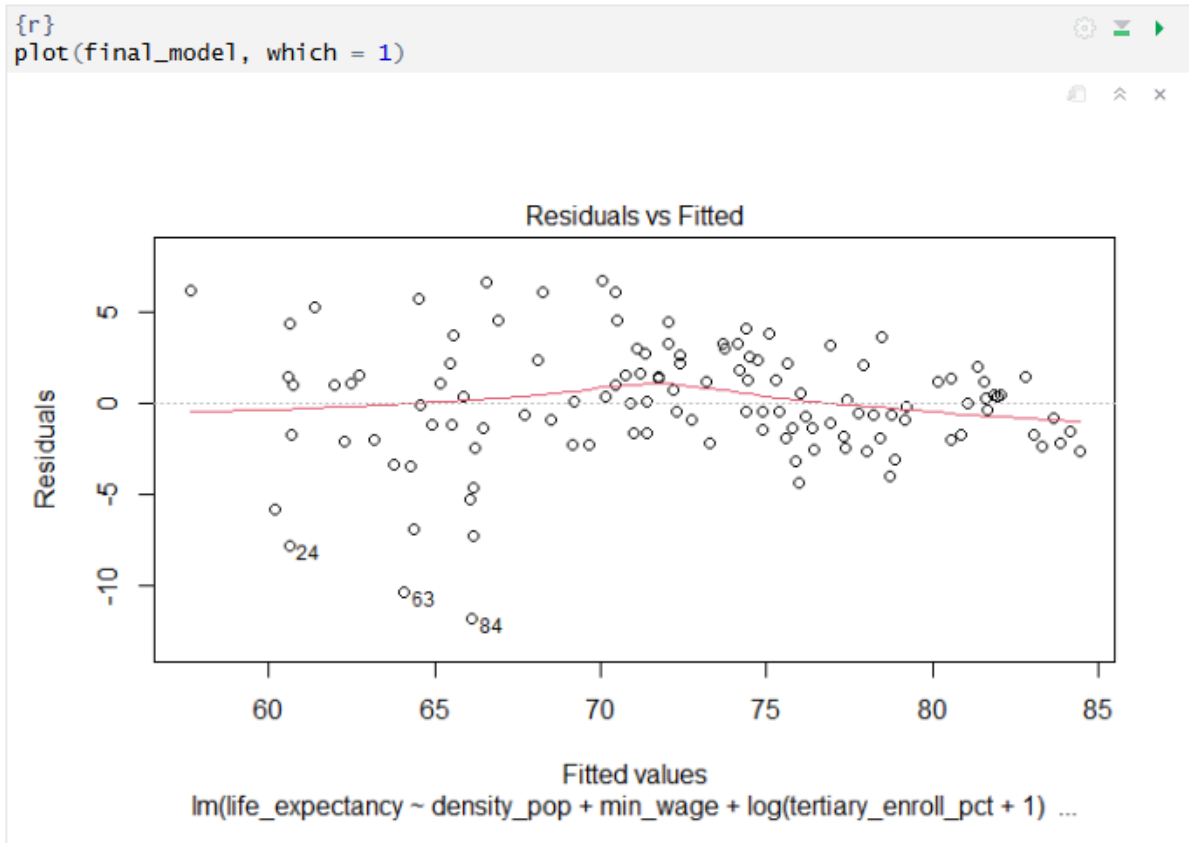
Predictors vs. fitted plot after transformation

```
{r}  
final_model <- update(best_model2, . ~ . + log(phys_per_1k + 1) - phys_per_1k + log  
(agri_land_pct + 1) - agri_land_pct, data = world_model_data)  
residualPlots(final_model)
```

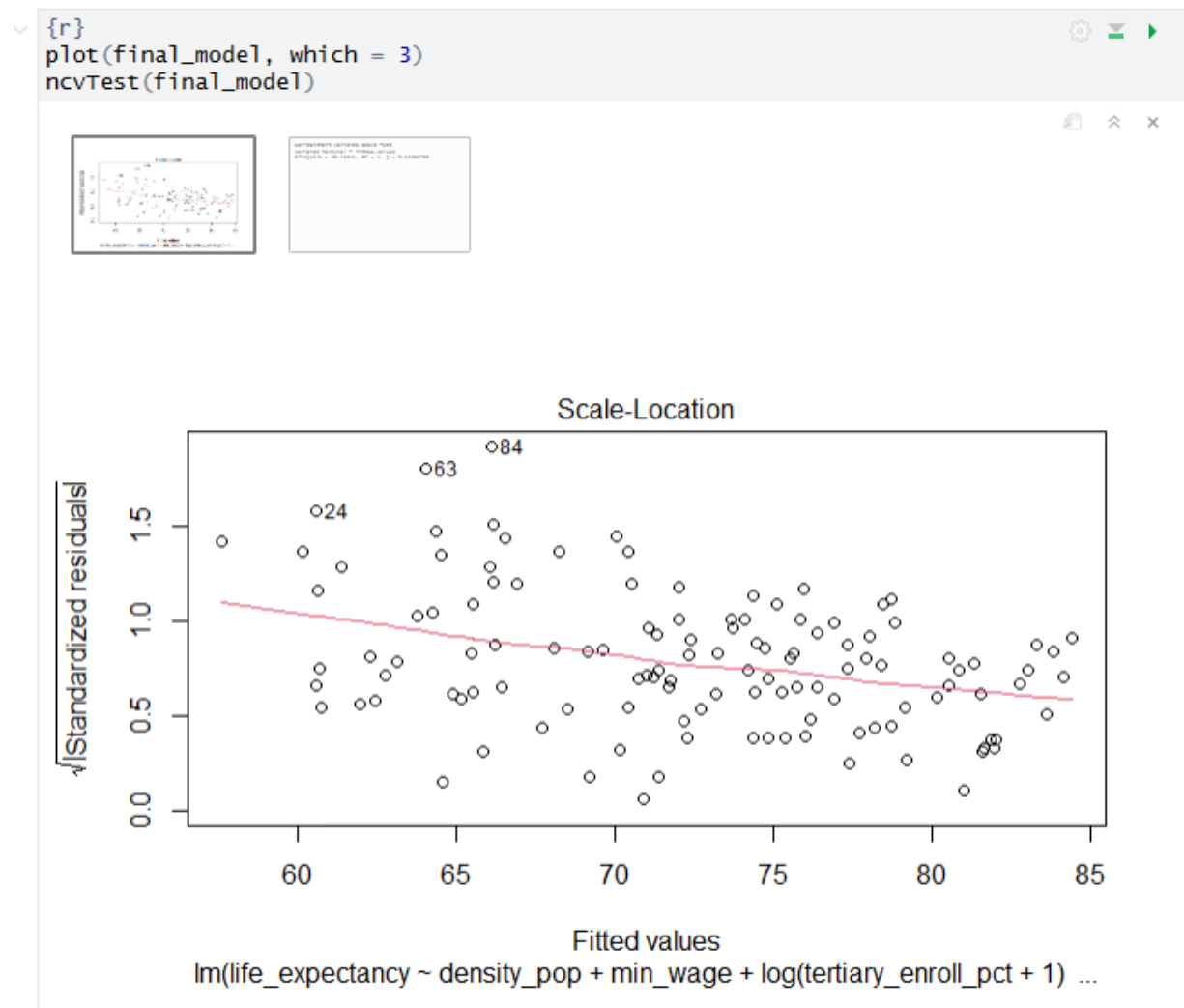
summary
AIC: 1000.000
BIC: 1000.000
Log Likelihood: -500.000
Number of Parameters: 10
Number of Observations: 100
R-squared: 0.999
Adjusted R-squared: 0.999
F-statistic: 1000.000
P-value: 0.000



New residuals vs. fitted plot after transformation

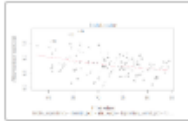


Scale-location plot



Non-constant variance test

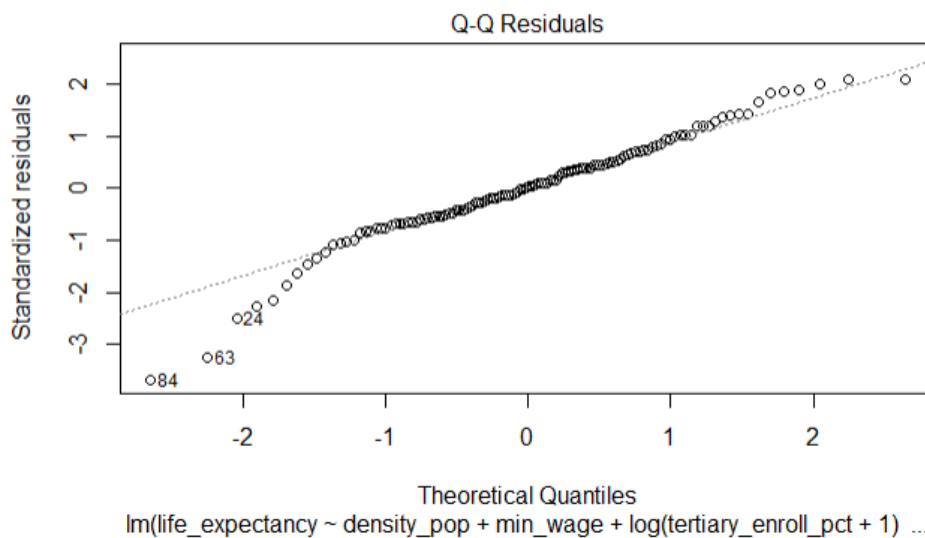
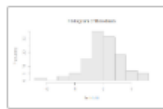
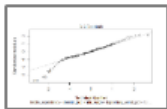
```
{r}  
plot(final_model, which = 3)  
ncvTest(final_model)
```



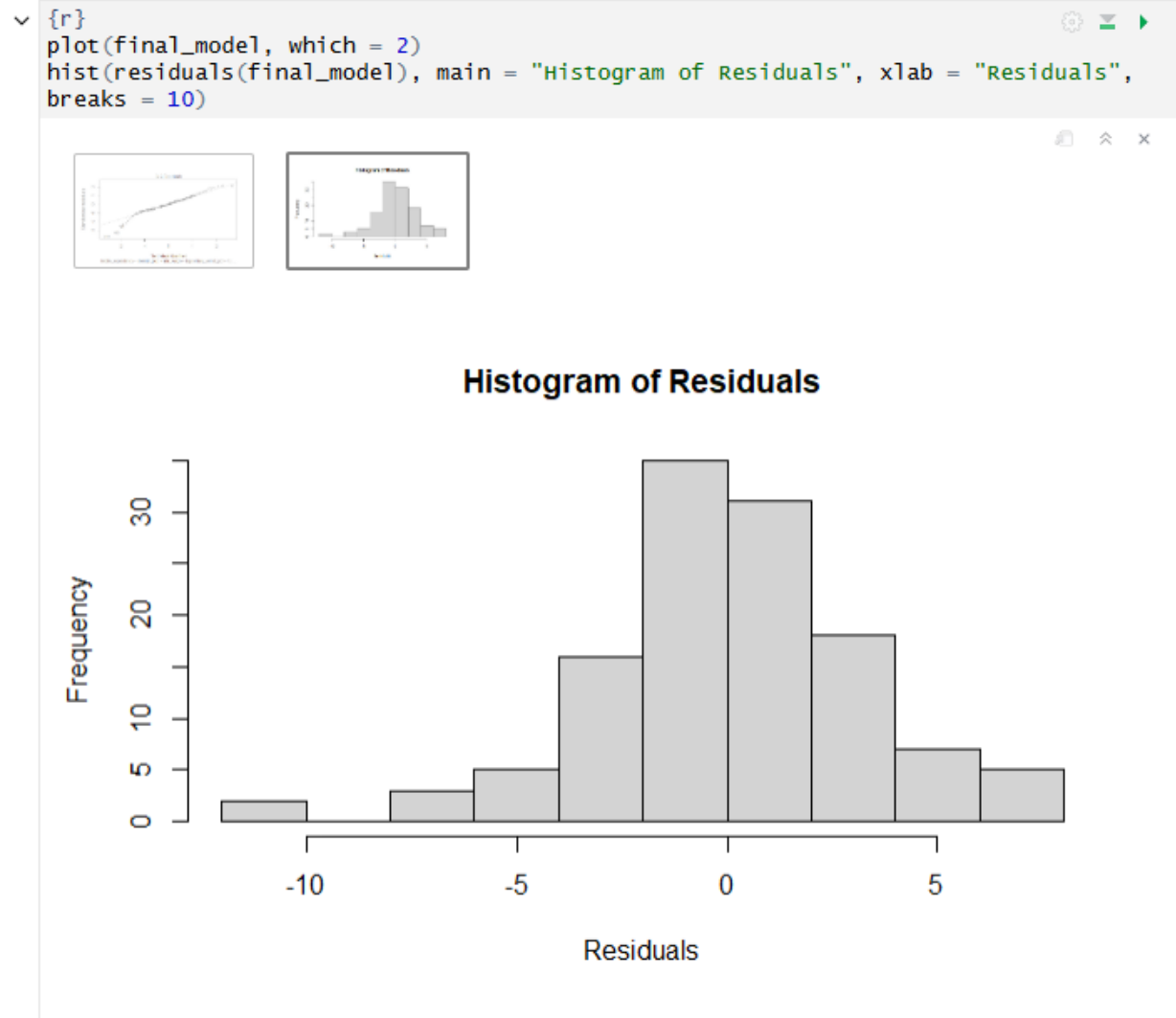
Non-constant Variance Score Test
variance formula: ~ fitted.values
chisquare = 28.46641, Df = 1, p = 9.5338e-08

Q-Q plot

```
{r}  
plot(final_model, which = 2)  
hist(residuals(final_model), main = "Histogram of Residuals", xlab = "Residuals",  
breaks = 10)
```

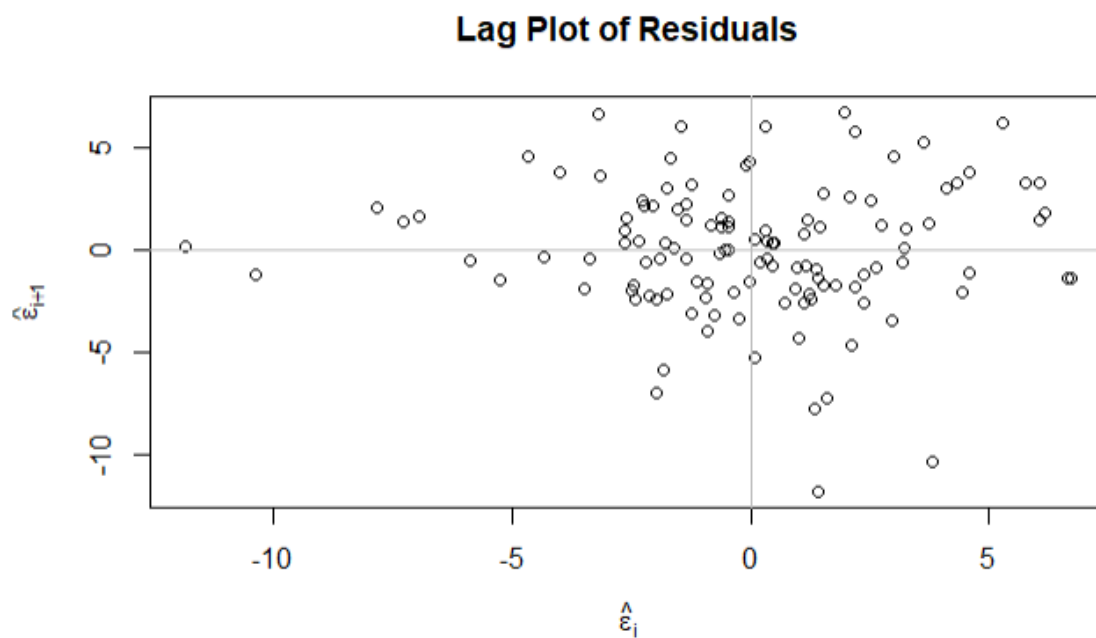
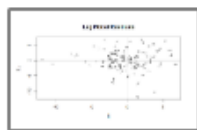


Histogram of residuals



Lag plot of residuals

```
{r}  
#check for serial correlation using lag plot and Durbin-watson test  
n <- nrow(world_model_data)  
plot(tail(residuals(final_model), n - 1) ~ head(residuals(final_model), n - 1), xlab  
= expression(hat(epsilon)[i]), ylab = expression(hat(epsilon)[i+1]), main = "Lag  
Plot of Residuals")  
  
abline(h= 0 , v = 0, col = grey(0.75))  
  
dwtest(final_model)
```



Durbin-Watson test

```
{r}
#check for serial correlation using lag plot and Durbin-watson test
n <- nrow(world_model_data)
plot(tail(residuals(final_model), n - 1) ~ head(residuals(final_model), n - 1), xlab
= expression(hat(epsilon)[i]), ylab = expression(hat(epsilon)[i+1]), main = "Lag
Plot of Residuals")

abline(h= 0 , v = 0, col = grey(0.75))

dwtest(final_model)
```



The image shows two side-by-side plots. The left plot is a 'Lag Plot of Residuals' with points scattered around a horizontal line at y=0. The right plot is the output of the 'dwtest' function, showing a DW statistic of 2.02 and a p-value of 0.5386.

Durbin-watson test

data: final_model
DW = 2.02, p-value = 0.5386
alternative hypothesis: true autocorrelation is greater than 0

Interpretation

Final model in interpretation

```
{r}
summary(final_model)

Call:
lm(formula = life_expectancy ~ density_pop + min_wage + log(tertiary_enroll_pct +
  1) + log(phys_per_1k + 1) + log(agri_land_pct + 1), data = world_model_data)

Residuals:
    Min       1Q   Median       3Q      Max
-11.8368  -1.7810   0.0437   1.9138   6.7247

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    54.7607     2.0931  26.162 < 2e-16 ***
density_pop      1.0119     0.2585   3.915 0.000153 ***
min_wage        3.1552     0.5664   5.571 1.67e-07 ***
log(tertiary_enroll_pct + 1)  3.8210     0.5718   6.682 8.61e-10 ***
log(phys_per_1k + 1)    2.4246     0.9643   2.514 0.013293 *
log(agri_land_pct + 1)  -1.2570     0.4757  -2.643 0.009361 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.269 on 116 degrees of freedom
Multiple R-squared:  0.8111,    Adjusted R-squared:  0.8029
F-statistic: 99.59 on 5 and 116 DF,  p-value: < 2.2e-16
```

$$\widehat{\text{Life_Exp}} = 60.5 + 0.885(\text{density_pop}) - 0.908(\text{agri_land_pct}) + 3.31(\text{min_wage}) + 1.62(\text{phys_per_1k}) + 2.58(\text{tertiary_enroll_pct})$$

Prediction

Confidence interval of predictions

```
{r}
new_data <- data.frame(
  density_pop = log(c(100, 500, 1000) + 1),
  agri_land_pct = log(c(20, 50, 80) + 1),
  min_wage = log(c(5, 10, 15) + 1),
  phys_per_1k = log(c(1, 3, 5) + 1),
  tertiary_enroll_pct = log(c(30, 60, 90) + 1)
)

predictions <- predict(final_model, newdata = new_data, interval = "prediction", level = 0.95)
predictions

      fit      lwr      upr
1 70.29479 63.27418 77.31540
2 74.95329 67.72693 82.17965
3 77.39158 70.00858 84.77459
```